

The work function of three photosensitive materials used to build photoelectric devices are given as : Sodium (2.75 eV), copper (4.65 eV) and gold (5.1 eV). Which of the following statements is correct. (The frequency of visible light lies in the range 4×10^{14} Hz to 8×10^{14} Hz) ?

A Devices built by copper and gold can operate with visible light

7.65% marked this

Devices built using sodium can operate with ultraviolet light

B Correct

71.98% marked this

All the devices can operate with infrared light

C 11.07% marked this

All the devices can operate with visible light

D 9.30% marked this

TIME TAKEN

Others: 201 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Medium

CHAPTER

Dual Nature of Matter

TOPIC

Photoelectric Effect

DIDN'T ATTEMPT !

28.48%

GOT IT RIGHT ✔

71.98%

GOT IT WRONG ✖

28.02%

SOLUTION

▶ Video Solution

Given, work function are as,

$$W_{Na} = 2.75 \text{ eV}$$

$$W_{Cu} = 4.65 \text{ eV}$$

$$\text{and } W_{Au} = 5.1 \text{ eV}$$

The frequency range of visible light 4×10^{14} Hz to 8×10^{14} Hz, so the maximum energy of photon and minimum energy of photon in visible light.

$$E_{\max} = \frac{6.63 \times 10^{-34} \times 8 \times 10^{14}}{1.6 \times 10^{-19}} \quad (\because E = h\nu)$$

$$= 331 \text{ eV}$$

$$\text{Similarly, } E_{\min} = \frac{6.63 \times 10^{-34} \times 4 \times 10^{14}}{1.6 \times 10^{-19}}$$

Now, comparing the work function of above mentioned metal with $E_{\max}$. We get only sodium device can give photoelectron.

Further sodium device can also work with photons of higher energy than 2.75 eV like x-rays, UV rays etc.

If E = energy, G = gravitational constant, I = impulse and M = mass, then dimensions of $\frac{GIM^2}{E^2}$ are same as that of.

Time

A

Correct

93.00% marked this

Mass

B

1.14% marked this

Length

C

2.63% marked this

Force

D

3.23% marked this

TIME TAKEN

Others: 147 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Easy

CHAPTER

Units and Dimensions

TOPIC

Dimensions

DIDN'T ATTEMPT



4.13%

GOT IT RIGHT



93.00%

GOT IT WRONG



7.00%

SOLUTION

▶ Video Solution

$$\text{Dimensions of } E = [ML^2 T^{-2}]$$

$$\text{Dimensions of } G = [M^{-1} L^3 T^{-2}]$$

$$\text{Dimensions of } I = [MLT^{-1}]$$

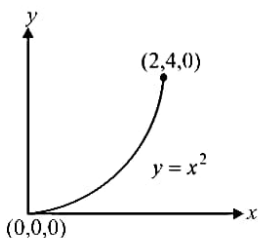
$$\text{And dimension of } M = [M]$$

$$\therefore \text{Dimensions of } \frac{GIM^2}{E^2} = \frac{[M^{-1} L^3 T^{-2}] [MLT^{-1}] [M^2]}{[ML^2 T^{-2}]^2}$$

$$= [T]$$

= Dimensions of time

A force $\vec{F} = (3xy - 5z)\hat{j} + 4z\hat{k}$ is applied to a particle. The work done by the force when the particle moves from point $(0, 0, 0)$ to point $(2, 4, 0)$ as shown in the figure is



A $\frac{280}{5}$ J
1.67% marked this

B $\frac{140}{5}$ J
1.08% marked this

C $\frac{232}{5}$ J
2.12% marked this

D $\frac{192}{5}$ J
Correct
95.13% marked this

TIME TAKEN

Others: 203 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Hard

CHAPTER

Work Power Energy

TOPIC

Work done

DIDN'T ATTEMPT

36.54%

GOT IT RIGHT

95.13%

GOT IT WRONG

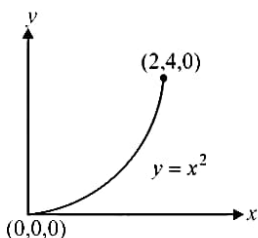
4.87%

SOLUTION

[▶ Video Solution](#)

Given that

$$\text{force } \vec{F} = (3xy - 5z)\hat{j} + 4z\hat{k}$$

Initial point $(0, 0, 0)$ Final point $(2, 4, 0)$ 

Work done by the variable force is, $W = \int \vec{F} \cdot d\vec{r}$.

$$\vec{F} = (3xy - 5z)\hat{j} + 4z\hat{k}$$

$$= (3xy)\hat{j}$$

$$y = x^2$$

$$dy = 2x dx$$

$$dw = F_x dx + F_y dy$$

$$= 0 + (3xy)dy$$

$$= 3x \times x^2 \times 2x dx$$

$$w = 6 \int_0^2 x^4 dx = 6 \cdot \left(\frac{x^5}{5} \right)_0^2 = \frac{192}{5} \text{ J}$$

A soap bubble (surface tension = T) is charged to a maximum surface density of charge = σ . When it is just going to burst. Its radius R is given by:

A $R = \frac{\sigma^2}{8\varepsilon_0 T}$

6.90% marked this

B $R = 8\varepsilon_0 \frac{T}{\sigma^2}$

Correct

75.99% marked this

C $R = \frac{\sigma}{\sqrt{8\varepsilon_0 T}}$

9.02% marked this

D $R = \frac{\sqrt{8\varepsilon_0 T}}{\sigma}$

8.09% marked this

EVALUATION

Not Answered

TIME TAKEN

Others: 120 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

Mechanical Properties of Fluids

TOPIC

Surface tension

DIDN'T ATTEMPT

53.44%

GOT IT RIGHT

75.99%

GOT IT WRONG

24.01%

SOLUTION

▶ Video Solution

The pressure due to surface tension = $\frac{4T}{R}$

The pressure due to electrostatic forces = $\frac{\sigma^2}{2\varepsilon_0}$

Just before the bubble bursts.

$$\frac{4T}{R} = \frac{\sigma^2}{2\varepsilon_0} \quad \text{or} \quad R = \frac{8T\varepsilon_0}{\sigma^2}$$

A long straight wire of radius a carries a steady current i . The current is uniformly distributed across its cross section. The ratio of the magnetic field at $a/2$ and $2a$ is,

- A** $1/2$
5.19% marked this
- B** $1/4$
6.61% marked this
- C** 4
23.22% marked this
- D** 1
Correct
64.97% marked this

TIME TAKEN

Others: 87 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Easy

CHAPTER

Magnetic Effects of Current

TOPIC

Magnetic Field

DIDN'T ATTEMPT 

9.23%

GOT IT RIGHT 

64.97%

GOT IT WRONG 

35.03%

SOLUTION

[▶ Video Solution](#)

Uniform current is flowing. Current enclosed in the first amperean path is,

$$\frac{I \cdot \pi r_1^2}{\pi R^2} = \frac{I r_1^2}{R^2}$$

$$\therefore B = \frac{\mu_0 \times \text{current}}{\text{path}} = \frac{\mu_0 \cdot I r_1^2}{2\pi r_1 R^2} = \frac{\mu_0 I r_1}{2\pi R^2}$$

Magnetic induction at a distance,

$$r_2 = \frac{\mu_0 I}{2\pi r_2}$$

$$\therefore \frac{B_1}{B_2} = \frac{r_1 r_2}{R^2} = \frac{\frac{a}{2} \cdot 2a}{a^2} = 1.$$

The primary and secondary coils of a transformer have 50 and 1500 turns respectively. If the magnetic flux ϕ linked with the primary coil is given by $\phi = \phi_0 + 4t$, where ϕ is in webers, t is time in second and ϕ_0 is a constant, the output voltage across the secondary coil is equal to 60 k V, then k is----

A 5
0.97% marked this

B 4
0.94% marked this

C 3
0.56% marked this

D 2
Correct
97.53% marked this

EVALUATION

TIME TAKEN

Others: 120 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Hard

CHAPTER

Alternating Current

TOPIC

Transformers

DIDN'T ATTEMPT 

22.84%

GOT IT RIGHT 

97.53%

GOT IT WRONG 

2.47%

SOLUTION

[▶ Video Solution](#)

Given : No. of turns across primary coil, $N_p = 50$
 Number of turns across secondary coil, $N_s = 1500$
 Magnetic flux linked with primary coil, $\phi = \phi_0 + 4t$

$\therefore$ Voltage across the primary coil,

$$V_p = \frac{d\phi}{dt} = \frac{d}{dt} (\phi_0 + 4t) = 4 \text{ volt}$$

$$\text{Also, } \frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$\therefore V_s = \left(\frac{1500}{50} \right) \times 4 = 120 \text{ V}$$

$$\Rightarrow k = 2$$

A solid sphere of mass M and radius R has a spherical cavity of radius $\frac{R}{2}$ such that the centre of cavity is at distance $R/2$ from the centre of the sphere. A point mass m is placed inside the cavity at a distance $R/4$ from the centre of sphere. The gravitational pull between the sphere and the point mass m is:

A $\frac{11GMm}{R^2}$
4.88% marked this

B $\frac{14GMm}{R^2}$
25.11% marked this

C $\frac{GMm}{2R^2}$
Correct
66.98% marked this

D $\frac{GMm}{R^2}$
3.03% marked this

TIME TAKEN

Others: 153 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Hard

CHAPTER

Electrostatics

TOPIC

Electric Field and Electric Field Lines

DIDN'T ATTEMPT

43.45%

GOT IT RIGHT

66.98%

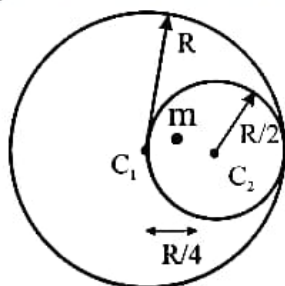
GOT IT WRONG

33.02%

SOLUTION

[▶ Video Solution](#)

To calculate the force of attraction on the point mass m , we should calculate the gravitational field inside the cavity and then force on m .



The gravitational field inside the cavity can be obtained by modifying Gauss law for gravitation, so,

$$\vec{E} = -\frac{4\pi G\rho}{3} \vec{C_1C_2} = -\frac{2\pi G\rho}{3} \vec{R} = -\frac{2\pi G}{3} \frac{M}{\frac{4}{3}\pi R^3} \vec{R} = -\frac{GM}{2R^2} \hat{R}$$

So,

$$F = mE = \frac{GMm}{2R^2}$$

Statement - 1: Two tuning forks having frequency 410 Hz and 524 Hz are kept close and made to vibrate. Beats will not be heard.

Statement - 2: Sound waves superimpose only when the frequencies of superposing waves are equal or nearly equal.

- A** Statement - 1 is True, Statement - 2 is True; Statement - 2 is a correct explanation for Statement - 1.

42.85% marked this

- B** Statement - 1 is True, Statement - 2 is True; Statement - 2 is NOT a correct explanation for Statement - 1.

2.44% marked this

- C** Statement - 1 is True, Statement - 2 is False.

Correct

26.48% marked this

- D** Statement - 1 is False, Statement - 2 is True.

28.23% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

90 sec

Others: 22 sec

DIFFICULTY

Easy

CHAPTER

Waves and Sound

TOPIC

Interference and Superposition of Waves

DIDN'T ATTEMPT



33.71%

GOT IT RIGHT



26.48%

GOT IT WRONG



73.52%

SOLUTION

▶ Video Solution

Number of beats = $524 - 410 = 114$ Due to persistence of hearing we can hear more than 10 beats per second. two waves of same frequencies and amplitude superimpose give stationary waves.

A dipole of dipole moment $\vec{p} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ is placed at the point $A(2, -3, 1)$. The electric potential due to this dipole at the point $B(4, -1, 0)$ is equal to (All the parameters specified here are in SI units.),

A $2 \times 10^9 \text{ V}$
8.81% marked this

B $-2 \times 10^9 \text{ V}$
Correct
79.74% marked this

C $3 \times 10^9 \text{ V}$
5.67% marked this

D $-3 \times 10^9 \text{ V}$
5.78% marked this

TIME TAKEN

Others: 162 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Medium

CHAPTER

Electrostatics

TOPIC

Electric Dipole

DIDN'T ATTEMPT 

35.22%

GOT IT RIGHT 

79.74%

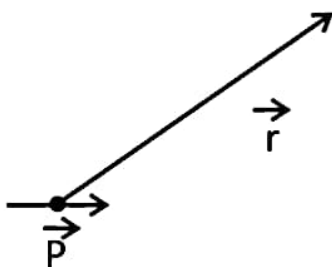
GOT IT WRONG 

20.26%

SOLUTION

[▶ Video Solution](#)

Let an electric dipole $\vec{p}$ is located at A .



The electric potential at a point due to a very small dipole is given by,

$$V = \frac{k(\vec{p} \cdot \vec{r})}{r^3}$$

$$\vec{r} = (4 - 2)\hat{i} + (-1 + 3)\hat{j} + (0 - 1)\hat{k} = 2\hat{i} + 2\hat{j} - \hat{k}$$

$$r = \sqrt{2^2 + 2^2 + (-1)^2} = \sqrt{9} = 3 \text{ m}$$

$$V_B = \frac{k(\vec{p} \cdot \vec{r})}{r^3} = \frac{k(2\hat{i} - 3\hat{j} + 4\hat{k}) \cdot (2\hat{i} + 2\hat{j} - \hat{k})}{3^3}$$

$$\Rightarrow V_B = \frac{9 \times 10^9 \times (-6)}{27} = -2 \times 10^9 \text{ V}$$

The temperature of 5 moles of a gas at constant volume is changed from 100°C to 120°C . The change in internal energy is 80 J. The total heat capacity of the gas at constant volume will be in J/K is :

A 8
0.99% marked this

B 4
Correct
29.73% marked this

C 0.8
68.70% marked this

D 0.4
0.59% marked this

DIFFICULTY
Medium

DIDN'T ATTEMPT 
9.72%

TIME TAKEN
Others: 37 sec

CHAPTER
Kinetic Theory of Gases

GOT IT RIGHT 
29.73%

TIME ALLOTTED
150 sec

TOPIC
Degree of Freedom and Specific Heat

GOT IT WRONG 
70.27%

SOLUTION

 Video Solution

$$n = 5 \text{ moles}, T_1 = 100^{\circ}\text{C}$$

$$T_2 = 120^{\circ}\text{C}, \Delta U = 80 \text{ J}$$

$$\text{Rise in temperature } \Delta t = 120 - 100 = 20^{\circ}\text{C}$$

$$\Delta U = ms\Delta t$$

$$\frac{80}{5} = 1 \times s \times 20$$

$$s = 0.8 \text{ J}$$

$$\therefore \text{ For 5 moles, } s = 0.8 \times 5 \text{ J/K}$$

$$= 4 \text{ J/K}$$

The energy of a photon is equal to the kinetic energy of a proton. The energy of the photon is E . Let λ_1 be the de-Broglie wavelength of the proton and λ_2 be the wavelength of the photon. The ratio $\frac{\lambda_1}{\lambda_2}$ is proportional to?

A E^0
13.84% marked this

B $E^{\frac{1}{2}}$
Correct
82.90% marked this

C E^{-1}
2.42% marked this

D E^{-2}
0.84% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 147 sec

120 sec

DIFFICULTY

CHAPTER

TOPIC

Easy

Dual Nature of Matter

Matter Waves

DIDN'T ATTEMPT GOT IT RIGHT GOT IT WRONG 

15.30%

82.90%

17.10%

SOLUTION

[▶ Video Solution](#)

Wavelength of the photon in terms of energy will be given by,

$$\lambda_2 = \frac{hc}{E}$$

De-Broglie of the proton will be,

$$\lambda_1 = \frac{h}{p} = \frac{h}{\sqrt{2m_p E}}$$

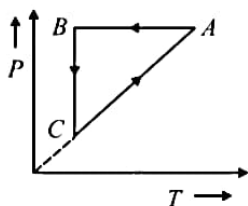
(where p is the momentum of the proton and m_p is the mass of the proton)

From above two equations,

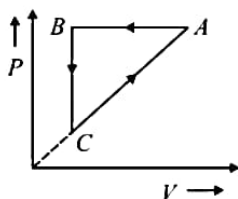
$$\frac{\lambda_1}{\lambda_2} = \frac{\frac{h}{\sqrt{2m_p E}}}{\frac{hc}{E}} = \frac{\sqrt{E}}{c\sqrt{2m_p}}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} \propto E^{\frac{1}{2}}$$

A cyclic process $ABCA$ is shown in $P - T$ diagram. What would be its $P - V$ graph:

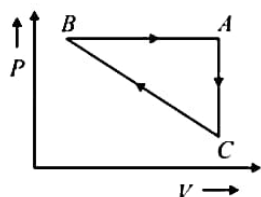


A



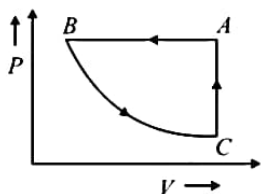
191% marked this

B



5.17% marked this

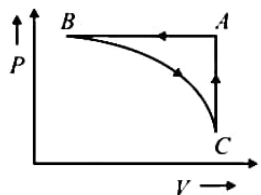
C



Correct

88.31% marked this

D



4.61% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 118 sec

150 sec

DIFFICULTY

CHAPTER

TOPIC

Medium

Thermodynamics

Thermodynamic Processes

DIDN'T ATTEMPT

GOT IT RIGHT

GOT IT WRONG

8.21%

88.31%

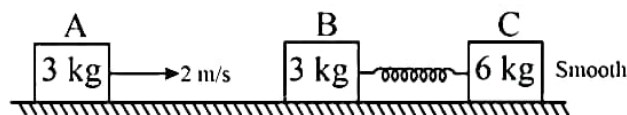
11.69%

SOLUTION

[▶ Video Solution](#)

$A \rightarrow B \Rightarrow$ constant pressure(linear)
 $B \rightarrow C \Rightarrow$ constant temperature(linear)
 $C \rightarrow A \Rightarrow$ constant volume(hyperbola)

For the shown situation, if a collision between blocks A and B is perfectly elastic, then find the maximum energy (in joules) stored in the spring.



4

A

Correct

76.70% marked this

2

B

10.51% marked this

1

C

2.12% marked this

3

D

10.66% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 119 sec

150 sec

DIFFICULTY

Medium

CHAPTER

Center of Mass Momentum and Collision

TOPIC

Collisions

DIDN'T ATTEMPT



33.28%

GOT IT RIGHT



76.70%

GOT IT WRONG



23.30%

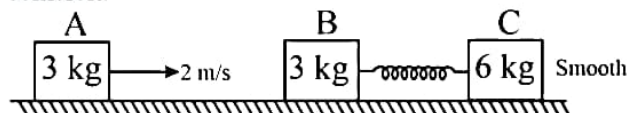
SOLUTION

[▶ Video Solution](#)

According to the conservation of linear momentum,

$$\vec{P}_i = \vec{P}_f,$$

where P_i is the initial momentum of the system and P_f is the momentum after collision.



Spring has maximum energy at the time of maximum compression. And at maximum compression, velocity of both the blocks should be equal. Applying momentum conservation,

$$P_i = P_f$$

$$3 \times 2 = (3 + 6) \times V$$

where, V is the velocity of both the blocks after collision.

$$V = \frac{2}{3} \text{ m s}^{-1}$$

Now, applying conservation of energy,

Change in kinetic energy = Potential energy gained

$$\Delta KE = \frac{1}{2} Kx^2 = U_{sp}$$

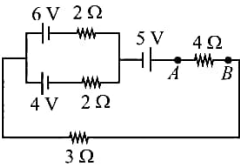
$$\Rightarrow \frac{1}{2} \times 3 \times \left(2\right)^2 - \left[\frac{1}{2} \left(3\right) \left(\frac{2}{3}\right)^2 + \frac{1}{2} \left(6\right) \left(\frac{2}{3}\right)^2 \right] = U_{\max}.$$

On solving,

$$U_{\max} = 4 \text{ J.}$$

Here, maximum energy stored in the spring is 4 J.

In the circuit shown (figure), calculate the current through 6 V battery.



- A

(1/4) A

5.97% marked this
- B

(1/8) A

4.59% marked this
- C

(1/2) A

Correct

72.71% marked this
- D

None of these

16.74% marked this

EVALUATION

DIFFICULTY
Easy

DIDN'T ATTEMPT 
29.24%

TIME TAKEN

Others: 152 sec

CHAPTER
Current Electricity

GOT IT RIGHT 
72.71%

TIME ALLOTTED

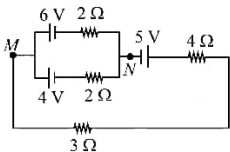
120 sec

TOPIC
Electric Cell or Battery

GOT IT WRONG 
27.29%

SOLUTION

[▶ Video Solution](#)



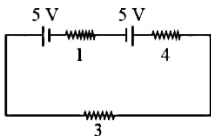
Equivalent EMF between points M and N ,

$$\epsilon_{eq} = \frac{\frac{\epsilon_1}{R_1} + \frac{\epsilon_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{\frac{6}{2} + \frac{4}{2}}{\frac{1}{2} + \frac{1}{2}}$$

$$\epsilon_{eq} = \frac{3+2}{1} = 5 \text{ V}$$

Equivalent resistance between M and N ,

$$\frac{1}{R} = \frac{1}{2} + \frac{1}{2} \Rightarrow R = 1 \, \Omega$$



Equivalent emf of circuit,

$$\epsilon_{eq} = 5 - 5 = 0$$

So, there will be no current in main circuit.

Current will only flow in small loop in the upper circuit.

Applying KVL in small loop,

$$6 - 4 - 4i = 0$$

Hence, current through 6 V battery is $\frac{1}{2}$ A.

A diffraction pattern is obtained using a beam of red light. What happens if the red light is replaced by blue light?

A Correct 82.86% marked this

B 13.58% marked this

C 1.04% marked this

D 2.52% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 56 sec

120 sec

DIFFICULTY

CHAPTER

TOPIC

Medium

Wave Optics

Diffraction of Light

DIDN'T ATTEMPT ○GOT IT RIGHT ✔GOT IT WRONG ✖

27.68%

82.86%

17.14%

SOLUTION

[▶ Video Solution](#)

The linear width of the central maxima in diffraction pattern is given by $\frac{2D\lambda}{a}$

where λ = Wavelength

a = Size of the slit

D = Distance between slit and screen.

As $\lambda_{\text{Blue}} < \lambda_{\text{Red}}$ and width of diffraction bands is directly proportional to λ .

Hence, when the red light is replaced by blue light, the wavelength decreases which means that the fringe's width decrease and become narrower and crowded together.

A screw gauge gives the following reading when used to measure the diameter of a wire.

Main scale reading : 0 mm

Circular scale reading: 52 divisions

Given that 1 mm on the main scale corresponds to 100 divisions of the circular scale.

The diameter of the wire from the above data is

0.052 cm

A

Correct

93.52% marked this

0.026 cm

B

0.59% marked this

0.005 cm

C

2.37% marked this

0.52 cm

D

3.52% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 83 sec

120 sec

DIFFICULTY

CHAPTER

TOPIC

Medium

Experimental Physics

Experimental Physics

DIDN'T ATTEMPT

GOT IT RIGHT

GOT IT WRONG

27.03%

93.52%

6.48%

SOLUTION

▶ Video Solution

Diameter of wire,

$$d = \text{MSR} + \text{CSR} \times \text{LC}$$

$$= 0 + 52 \times \frac{1}{100}$$

$$= 0.52 \text{ mm} = 0.052 \text{ cm.}$$

/

The space has electromagnetic field which varies with time whose variation is given as:

$$\vec{B} = \begin{cases} -B_0 \hat{k} & \text{if } 0 \leq t \leq \frac{\pi m}{qB_0} \\ -B_0 \hat{j} & \text{if } \frac{\pi m}{qB_0} \leq t \leq \frac{2\pi m}{qB_0} \\ 0 & \text{if } \frac{2\pi m}{qB_0} \leq t < \infty \end{cases} \text{ and } \vec{E} = \begin{cases} 0 & \text{if } 0 \leq t \leq \frac{2\pi m}{qB_0} \\ -E_0 \hat{k} & \text{if } \frac{2\pi m}{qB_0} \leq t \leq \infty \end{cases}$$

A charge particle having mass m and positive charge q is given velocity $v_0 \hat{i}$ at origin at $t = 0$ sec.

The coordinate of point on xy plane when it again passes through xy plane for the first time is of the form $\left(\frac{xmv_0}{q} \sqrt{\frac{v_0}{E_0 B_0}}, y \frac{mv_0}{qB_0}, 0 \right)$.

Find $x + y$?

- A** 5
5.04% marked this
- B** 6
5.40% marked this
- C** 3
13.13% marked this
- D** 4
Correct
76.44% marked this

EVALUATION

Not Answered

TIME TAKEN

You:
Others: 239 sec

TIME ALLOTTED

210 sec

DIFFICULTY

Hard

CHAPTER

Magnetic Effects of Current

TOPIC

Motion of Charged Particle in Magnetic Field

DIDN'T ATTEMPT

75.50%

GOT IT RIGHT

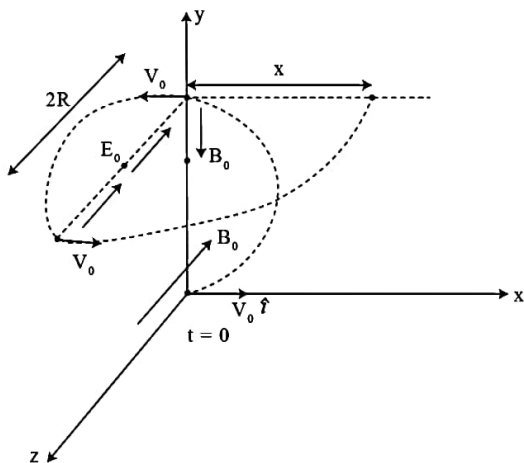
76.44%

GOT IT WRONG

23.56%

SOLUTION

[▶ Video Solution](#)



[$OA \rightarrow$ circular path]

[$AB \rightarrow$ circular path]

[$BC \rightarrow$ parabolic path]

From B to C ,

$$\text{acceleration} = \frac{qE_0}{m}$$

$$t = \sqrt{\frac{2\lambda}{a}} = \sqrt{\frac{4m^2 V_0}{q^2 E_0 B_0}}$$

$$\therefore x = vt = \frac{2mv_0}{q} \sqrt{\frac{V_0}{E_0 B_0}}$$

Co-ordinate of pt. C

$$\Rightarrow \left(\frac{2mv_0}{q} \sqrt{\frac{V_0}{E_0 B_0}}, \frac{2mv_0}{qB_0}, 0 \right)$$

$$\therefore x = 2$$

$$y = 2$$

!

A wheel of moment of inertia 2.5 Kg-m^2 has an initial angular velocity of 40 rads^{-1} . A constant torque of 10 Nm acts on the wheel. The time during which the wheel is accelerated to 60 rads^{-1} is.....

A **6 s**
0.50% marked this

B **5 s**
Correct
98.09% marked this

C **4.5 s**
0.66% marked this

D **4 s**
0.76% marked this

EVALUATION

TIME TAKEN

Others: 81 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Easy

CHAPTER

Rotational Motion

TOPIC

Rotational kinematics

DIDN'T ATTEMPT

14.76%

GOT IT RIGHT

98.09%

GOT IT WRONG

1.91%

SOLUTION

Video Solution

Given, $MI = 2.5 \text{ kgm}^{-2}$

$W = 40 \text{ rad s}^{-1}$

$T = 10 \text{ Nm}$

As $T = I\alpha$

$10 = 2.5\alpha$

$\alpha = 4 \text{ rad s}^{-2}$

Now, $\omega = \omega_0 + \alpha t$

$60 = 40 + 4 \times t$

$20 = 4t$

$t = 5 \text{ s}$

A beaker contains water up to a height h_1 and kerosene of height h_2 above water so that the total height of (water + kerosene) is $(h_1 + h_2)$. Refractive index of water is μ_1 and that of kerosene is μ_2 . The apparent shift in the position of the bottom of the beaker when viewed from above is?

A $\left(1 - \frac{1}{\mu_1}\right)h_2 + \left(1 - \frac{1}{\mu_2}\right)h_1$

4.68% marked this

B $\left(1 + \frac{1}{\mu_1}\right)h_1 + \left(1 + \frac{1}{\mu_2}\right)h_2$

3.56% marked this

C $\left(1 - \frac{1}{\mu_1}\right)h_1 + \left(1 - \frac{1}{\mu_2}\right)h_2$

Correct

91.25% marked this

D $\left(1 + \frac{1}{\mu_1}\right)h_2 - \left(1 + \frac{1}{\mu_2}\right)h_1$

0.51% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 96 sec

150 sec

DIFFICULTY

Medium

CHAPTER

Ray Optics

TOPIC

Refraction of light from plane surface

DIDN'T ATTEMPT 

28.74%

GOT IT RIGHT 

91.25%

GOT IT WRONG 

8.75%

SOLUTION

[▶ Video Solution](#)

Apparent shift due to a transparent material of thickness h is give by,

$$\Delta h = \left(1 - \frac{1}{\mu}\right)h$$

$\therefore$ Apparent shift produced by water

$$\Delta h_1 = \left(1 - \frac{1}{\mu_1}\right)h_1$$

And apparent shift produced by kerosene

$$\Delta h_2 = \left(1 - \frac{1}{\mu_2}\right)h_2$$

$$\Delta h = \Delta h_1 + \Delta h_2 = \left(1 - \frac{1}{\mu_1}\right)h_1 + \left(1 - \frac{1}{\mu_2}\right)h_2$$

The escape velocity of an object from a planet is 16 km s^{-1} . If the escape velocity of the object from another planet having twice the density and three times the radius of the planet is $v\sqrt{2}\text{ ms}^{-1}$, then the value of v is ____

A

12

6.65% marked this

B

48

Correct

89.22% marked this

C

18

1.90% marked this

D

36

2.23% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 169 sec

120 sec

DIFFICULTY
MediumCHAPTER
GravitationTOPIC
Gravitational potential and Potential EnergyDIDN'T ATTEMPT 
12.27%GOT IT RIGHT 
89.22%GOT IT WRONG 
10.78%

SOLUTION

[▶ Video Solution](#)

Escape speed is given by

$$v = \sqrt{\frac{2GM}{R}}$$

$$= \sqrt{\frac{2G \times \frac{4}{3} \pi R^3 \times d}{R}} = (\because M = d \times V).$$

$$\Rightarrow v = \sqrt{\frac{8}{3} G \pi R^2 d}$$

Now given for a planet (lets say A)

$$v_1 = 16 = \sqrt{\frac{8}{3} G \pi R^2 d}$$

and for planet B,

$$v_2 = \sqrt{\frac{8}{3} G \pi (3R)^2 \times 2d}.$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{16}{v_2} = \sqrt{\frac{R^2 \times d}{(3R)^2 \times 2d}}$$

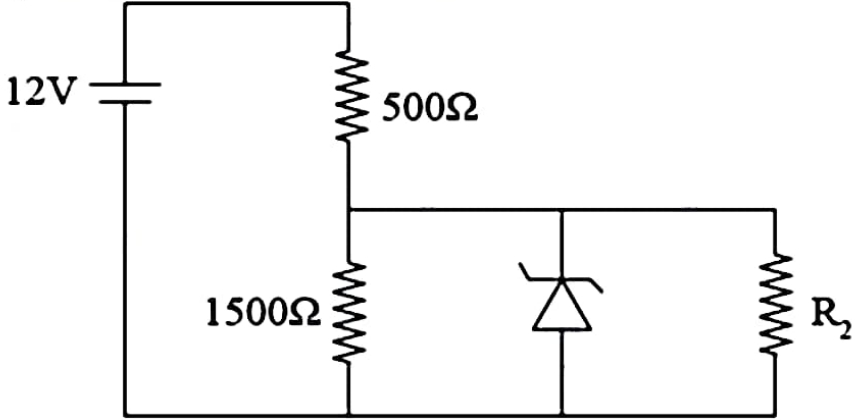
$$\Rightarrow \frac{16}{v_2} = \sqrt{\frac{1}{18}} = \frac{1}{3\sqrt{2}}$$

Hence, $v_2 = 48\sqrt{2}\text{ m/s} = v\sqrt{2}\text{ ms}^{-1}(\text{given})$

$$\therefore v = 48$$

/

A Zener diode of Zener break-down voltage 10 V is connected as shown in the figure. Current through Zener diode is



Your Answer

Correct Answer

0

TIME TAKEN

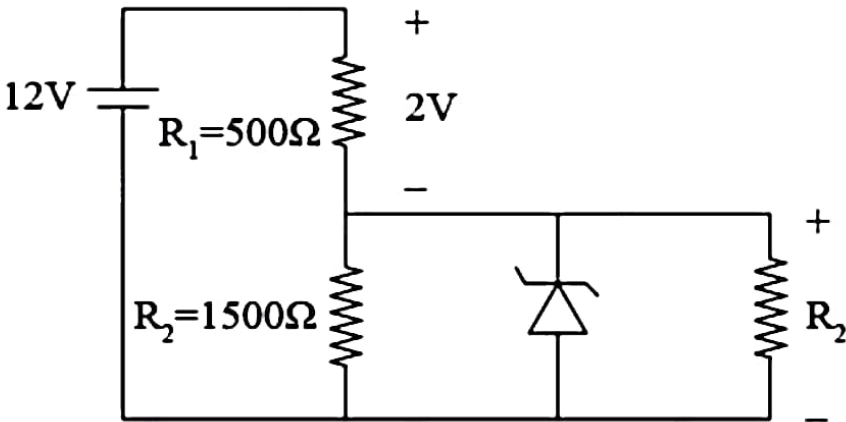
Others: 161 sec

TIME ALLOTTED

150 sec

DIFFICULTY
MediumCHAPTER
SemiconductorsTOPIC
DiodesDIDN'T ATTEMPT
50.45%GOT IT RIGHT
92.03%GOT IT WRONG
7.97%

SOLUTION

[▶ Video Solution](#)

If we consider break down in Zener diode, then potential across R_3 will be 10 V and R_1 will be 2 V.

So current in R_3 will be $i_3 = \frac{10}{1500} = \frac{2}{300}$ A

and current in R_1 will be $i_1 = \frac{2}{500}$ A

$\Rightarrow i_1 < i_3$, which is not possible

$\Rightarrow$ The potential difference across the Zener diode does not reach to break down voltage. So no current will flow through the reverse-biased Zener diode.

A solenoid of length 60 cm with 15 turns per cm and area of cross section $4 \times 10^{-3} \text{ m}^2$ completely surrounds another co-axial solenoid of same length and area of cross-section $2 \times 10^{-3} \text{ m}^2$ with 40 turns per cm. Mutual inductance of the system is _____ mH. (Nearest Integer)

Your Answer

Correct Answer

9

TIME TAKEN

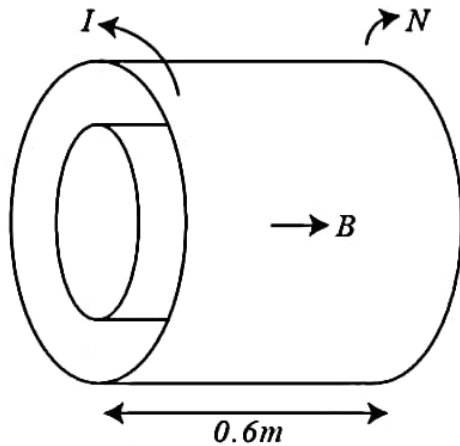
Others: 124 sec

TIME ALLOTTED

150 sec

DIFFICULTY
HardCHAPTER
Electromagnetic InductionTOPIC
Self and Mutual InductanceDIDN'T ATTEMPT 
55.29%GOT IT RIGHT 
49.85%GOT IT WRONG 
50.15%

SOLUTION

[▶ Video Solution](#)

$$N = 15/\text{cm} = 1500$$

$$B = \mu_0 N I$$

$$= \mu_0 (1500) I$$

$$\phi = (B) (2 \times 10^{-3}) (40 \times 60)$$

$$= \underbrace{\mu_0 (1500) (2 \times 10^{-3} \times 2.400)}_M I \propto I$$

$$M = 4\pi \times 10^{-7} \times 1.5 \times 10^3 \times 2 \times 2.4$$

$$= 9.04 \times 10^{-3} \text{ H}$$

Position (in m) of a particle moving on a straight line varies with time (in sec) as $x = \frac{t^3}{3} - 3t^2 + 8t + 4$ (m). Consider the motion of the particle from $t = 0$ to $t = 5$ sec. S_1 is the total distance travelled and S_2 is the distance travelled during retardation. If $\frac{S_1}{S_2} = \frac{(3\alpha+2)}{11}$ then find α .

Your Answer

Correct Answer

4

TIME TAKEN

Others: 192 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Easy

CHAPTER

Motion In One Dimension

TOPIC

Non-uniform Motion

DIDN'T ATTEMPT 

50.04%

GOT IT RIGHT 

65.04%

GOT IT WRONG 

34.96%

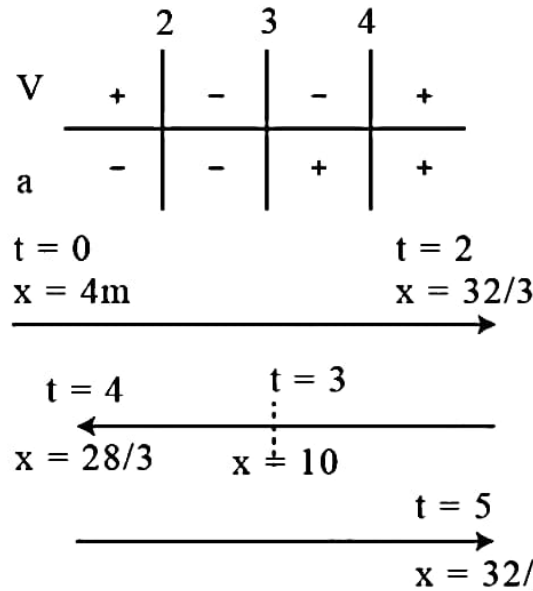
SOLUTION

[▶ Video Solution](#)

$$x = \frac{t^3}{3} - 3t^2 + 8t + 4$$

$$v = t^2 - 6t + 8 = (t - 2)(t - 4)$$

$$a = 2(t - 3)$$

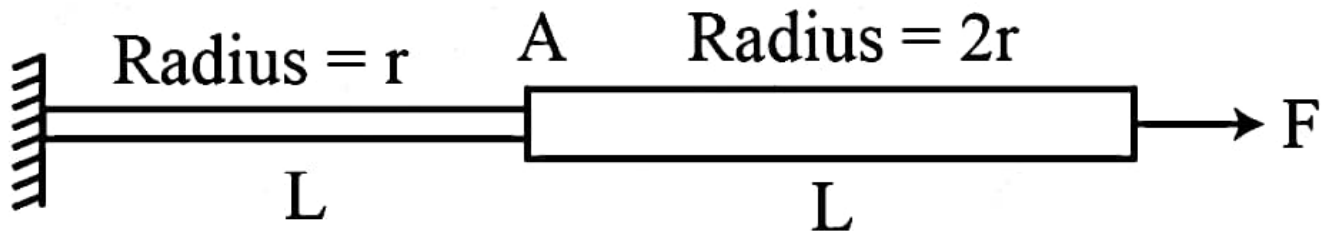


$$S_1 = \left(\frac{32}{3} - 4\right) + \left(\frac{32}{3} - \frac{28}{3}\right) + \left(\frac{32}{3} - \frac{28}{3}\right) = \frac{20}{3} + \frac{8}{3} = \frac{28}{3} \text{ m}$$

$$S_2 = \left(\frac{32}{3} - 4\right) + \left(10 - \frac{28}{3}\right) = \frac{20}{3} + \frac{2}{3} = \frac{22}{3} \text{ m}$$

$$\frac{S_1}{S_2} = \frac{28}{22} = \frac{14}{11} = \frac{3\alpha+2}{11} \Rightarrow \alpha = 4$$

Two steel wires of same length but radii r and $2r$ are connected together end to end and tied to a wall as shown. The force stretches the combination by 10 mm. How far does the midpoint A move. (in mm)



Your Answer

Correct Answer

8

EVALUATION

TIME TAKEN

Others: 122 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Medium

CHAPTER

Mechanical Properties of Solids

TOPIC

Young's Modulus and Breaking Stress

DIDN'T ATTEMPT

53.87%

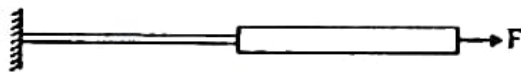
GOT IT RIGHT

84.10%

GOT IT WRONG

15.90%

SOLUTION

[▶ Video Solution](#)

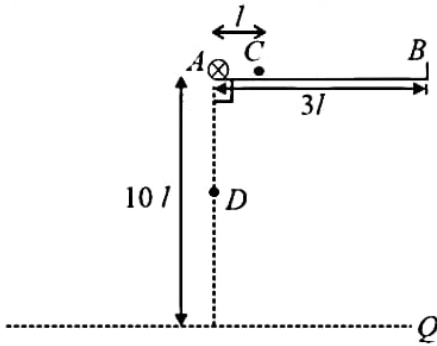
$$\frac{F}{A_1} = y \frac{\Delta \ell_1}{\ell}$$

$$\frac{F}{A_2} = y \frac{\Delta \ell_2}{\ell}$$

$$\Delta \ell_1 + \Delta \ell_2 = 10 \text{ mm}$$

$$\frac{F\ell}{A_1 y} + \frac{F\ell}{4 A_1 y} = 10 \text{ mm} \Rightarrow \frac{F\ell}{A_1 y} = 8 \text{ mm}$$

A uniform thin rod of mass 'm' and length ' $3l$ ' is released from rest from horizontal position as shown. When it passes through the vertical line AD, it gets broken at point 'C' due to some reason. Find the angle in radian rotated by rod BC till its centre of mass passes through the horizontal line PQ from the instant of breakage. ($g = 10 \text{ m/s}^2$) (hinge is smooth and there is no thrust on one part of rod due to the other part at the time of breaking.)



Your Answer

Correct Answer

4

TIME TAKEN

Others: 112 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Hard

CHAPTER

Rotational Motion

TOPIC

Angular momentum and Angular impulse

DIDN'T ATTEMPT

83.95%

GOT IT RIGHT

75.65%

GOT IT WRONG

24.35%

SOLUTION

[▶ Video Solution](#)

By energy conservation

$$mg \cdot \frac{3l}{2} = \frac{1}{2} \cdot \frac{m(3l)^2}{3} \cdot \omega^2$$

$$\omega = \sqrt{\frac{g}{l}}$$

And velocity of centre of mass of rod BC $V_{cm} = \sqrt{\frac{g}{l}} \cdot 2l$

$$V_{cm} = 2\sqrt{gl}$$

If time taken by centre of mass of rod BC from breaking position to line PQ is t.

$$8l = \frac{1}{2} \times g \times t^2$$

$$t = 4\sqrt{\frac{l}{g}}$$

$$\theta = \omega \cdot t = 4 \text{ radian}$$

The increasing order of Ag^+ ion concentration in

I. Saturated solution of AgCl

II. Saturated solution of AgI

III. $1\text{M Ag}(\text{NH}_3)_2^+$ in 0.1M NH_3

IV. $1\text{M Ag}(\text{CN})_2^-$ in 0.1M KCN

Given:

$$K_{\text{sp}} \text{ of } \text{AgCl} = 1.0 \times 10^{-10}$$

$$K_{\text{sp}} \text{ of } \text{AgI} = 1.0 \times 10^{-16}$$

$$K_{\text{d}} \text{ of } \text{Ag}(\text{NH}_3)_2^+ = 1.0 \times 10^{-8}$$

$$K_{\text{d}} \text{ of } \text{Ag}(\text{CN})_2^- = 1.0 \times 10^{-21}$$

A $I < II < III < IV$
1.11% marked this

B $IV < III < II < I$
9.30% marked this

C $IV < II < III < I$
Correct
67.42% marked this

D $IV < II < I < III$
22.18% marked this

TIME TAKEN

Others: 117 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Medium

CHAPTER

Ionic Equilibrium

TOPIC

Solubility

DIDN'T ATTEMPT

46.41%

GOT IT RIGHT

67.42%

GOT IT WRONG

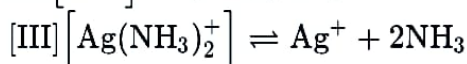
32.58%

SOLUTION

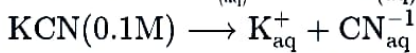
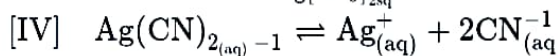
▶ Video Solution

$$[\text{I}] \left[\text{Ag}^+ \right] = \sqrt{K_{\text{sp}}(\text{AgCl})} = \sqrt{1 \times 10^{-10}} = 10^{-5}\text{M}$$

$$[\text{II}] \left[\text{Ag}^+ \right] = \sqrt{K_{\text{sp}}(\text{AgI})} = \sqrt{1 \times 10^{-16}} = 10^{-8}\text{M}$$



$$K_{\text{d}} = 1 \times 10^{-8} = \frac{[\text{Ag}^+][\text{NH}_3]^2}{[\text{Ag}(\text{NH}_3)_2^+]} = \frac{[\text{Ag}^+](0.1)^2}{1.0} \Rightarrow [\text{Ag}^+] = 1 \times 10^{-6}\text{M}$$



$$K_{\text{d}} = 1 \times 10^{-21} = \frac{[\text{Ag}^+][\text{CN}^-]^2}{[\text{Ag}(\text{CN})_2]^{-1}} = \frac{[\text{Ag}^+](0.1)^2}{1.0}$$

$$[\text{Ag}^+] = 1 \times 10^{-19}\text{M}$$

The electronic configurations for some neutral atoms are given below

(I) $1s^2, 2s^2 2p^6, 3s^2$

(II) $1s^2, 2s^2 2p^6, 3s^1$

(III) $1s^2, 2s^2 2p^6, 3s^2 3p^2$

(IV) $1s^2, 2s^2 2p^6, 3s^2 3p^3$

Which of these is expected to have the highest second ionisation enthalpy?

I

A

5.69% marked this

II

B

Correct

86.89% marked this

III

C

2.44% marked this

IV

D

4.97% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

60 sec

Others: 50 sec

DIFFICULTY

Easy

CHAPTER

Classification of Elements and
Periodicity in Properties

TOPIC

Periodic properties

DIDN'T ATTEMPT

2.74%

GOT IT RIGHT

86.89%

GOT IT WRONG

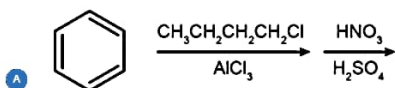
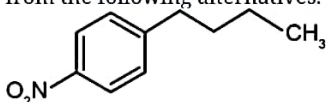
13.11%

SOLUTION

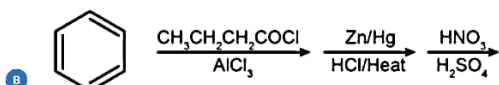
▶ Video Solution

B atom after losing outermost electron acquires noble gas configuration (stable configuration). It is difficult to remove the next electron from $B^+ (1s^2, 2s^2 2p^6)$ ion.

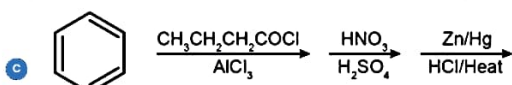
Identify the correct method for the synthesis of the compound shown below from the following alternatives.



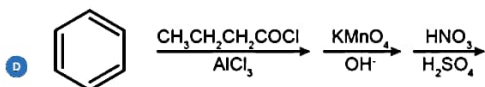
27.57% marked this



Correct
61.92% marked this



9.31% marked this



1.20% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

180 sec

Others: 65 sec

DIFFICULTY

Hard

CHAPTER

Hydrocarbons

TOPIC

Reactions of aromatic compounds

DIDN'T ATTEMPT 

16.16%

GOT IT RIGHT 

61.92%

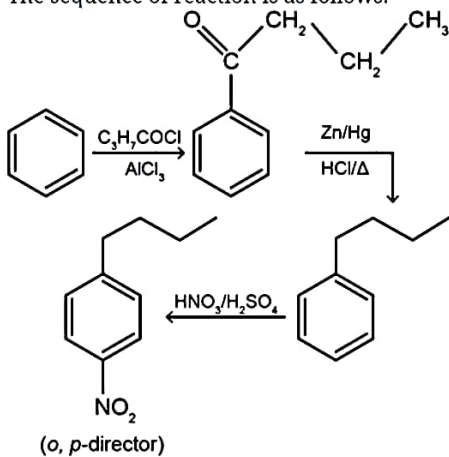
GOT IT WRONG 

38.08%

SOLUTION

[▶ Video Solution](#)

The sequence of reaction is as follows:



In the following sub questions, choose the correct answer from among the following possibilities and select correct code of your answer (Answer of 1, 2, 3 and 4 respectively)

1) The most stable low valent halide (1) GeCl_2 (2) SnCl_2 (3) PbCl_2 .

2) A non existing halide (1) SnCl_4 (2) PbCl_4 (3) PbI_4 .

3) A purely acidic oxide (1) PbO_2 (2) SnO_2 (3) SiO_2

4) Thermally most stable hydride (1) NH_3 (2) PH_3 (3) AsH_3 .

A 3, 2, 1, 3
4.29% marked this

B 1, 3, 3, 1
12.87% marked this

C 3, 3, 3, 1
Correct
81.66% marked this

D 1, 1, 1, 3
1.17% marked this

TIME TAKEN

Others: 97 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

p Block Elements (Group 15, 16, 17 & 18)

TOPIC

Nitrogen and Its Compounds

DIDN'T ATTEMPT

21.09%

GOT IT RIGHT

81.66%

GOT IT WRONG

18.34%

SOLUTION

[▶ Video Solution](#)

Stability $\text{Pb}^{+2} > \text{Sn}^{+2} > \text{Ge}^{+2}$

in PbI_4 , Pb^{+4} is less stable

Acidic nature :

$\text{CO}_2 > \text{SiO}_2 > \text{GeO}_2 > \text{SnO}_2 > \text{PbO}_2$

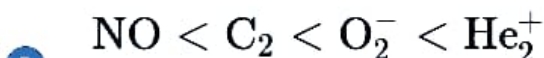
GeO_2 SnO_2 PbO_2 = amphoteric

B.E. = $\text{N} - 4 > \text{B} - 4 > \text{As} - 4$

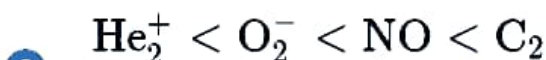
Which of the following is a correct order of bond order ?



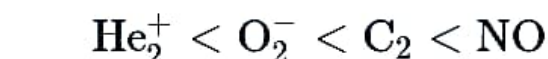
1.26% marked this



0.90% marked this



14.50% marked this



Correct

83.34% marked this

TIME TAKEN

Others: 108 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Easy

CHAPTER

Chemical Bonding and Molecular Structure

TOPIC

Molecular Orbital Theory

DIDN'T ATTEMPT

8.08%

GOT IT RIGHT

83.34%

GOT IT WRONG

16.66%

SOLUTION

▶ Video Solution

Bond order of $\text{He}_2^+ \rightarrow 0.5$

$\text{O}_2^- \rightarrow 1.5$

$\text{C}_2 \rightarrow 2$

$\text{NO} \rightarrow 2.5$

A sample of hydrogen atoms de-excites from 6th excited state to the ground state in one or more electronic transitions. If no spectral line is obtained in the Paschen and Brackett series, then the maximum number of spectral lines of different photon energies obtained will be -

A 21
7.63% marked this

B 15
24.20% marked this

C 10
Correct
55.23% marked this

D 7
12.94% marked this

EVALUATION

TIME TAKEN

Others: 104 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

Structure of Atom

TOPIC

Hydrogen spectrum

DIDN'T ATTEMPT

22.60%

GOT IT RIGHT

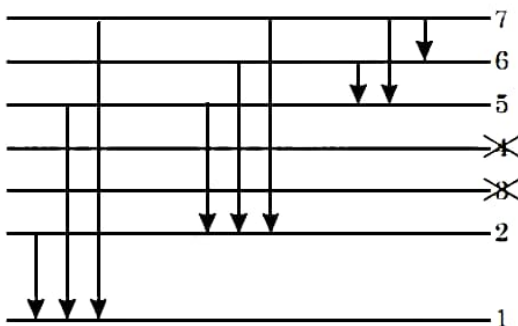
55.23%

GOT IT WRONG

44.77%

SOLUTION

▶ Video Solution



The correct statement is .

- A** BI_3 is the weakest Lewis acid among the boron halides
2.71% marked this
- B** there is minimum $\text{p}\pi - \text{p}\pi$ back bonding in BF_3
11.88% marked this
- C** BF_3 is the strongest Lewis acid among the other boron halides
7.89% marked this
- D** there is maximum $\text{p}\pi - \text{p}\pi$ back bonding in BF_3
Correct
77.52% marked this

EVALUATION

TIME TAKEN

Others: 45 sec

TIME ALLOTTED

60 sec

DIFFICULTY

Easy

CHAPTER

p Block Elements (Group 13 & 14)

TOPIC

Boron and Its Compounds

DIDN'T ATTEMPT

12.38%

GOT IT RIGHT

77.52%

GOT IT WRONG

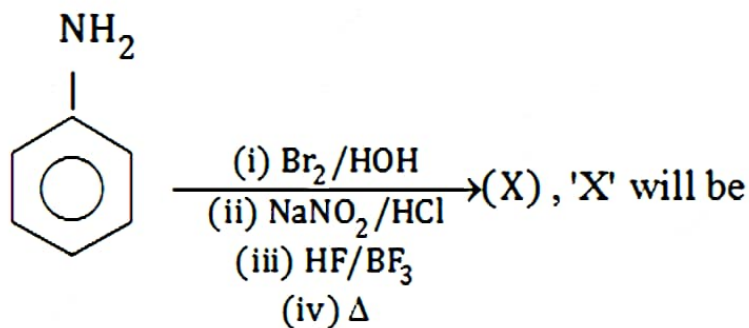
22.48%

SOLUTION

[▶ Video Solution](#)

There is significant $\text{p}\pi - \text{p}\pi$ back bonding in BF_3 due to almost similar sizes of 2p-orbitals of B and F.

In the reaction



- A** p-Bromofluorobenzene
8.80% marked this
- B** 2, 4, 6-Tribromofluorobenzene
Correct
83.70% marked this
- C** p-Bromoaniline
0.76% marked this
- D** 1, 3, 5-Tribromobenzene
6.73% marked this

TIME TAKEN

Others: 69 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Hard

CHAPTER

Amines

TOPIC

Preparation and Properties of Amines

DIDN'T ATTEMPT 🟡

20.03%

GOT IT RIGHT ✅

83.70%

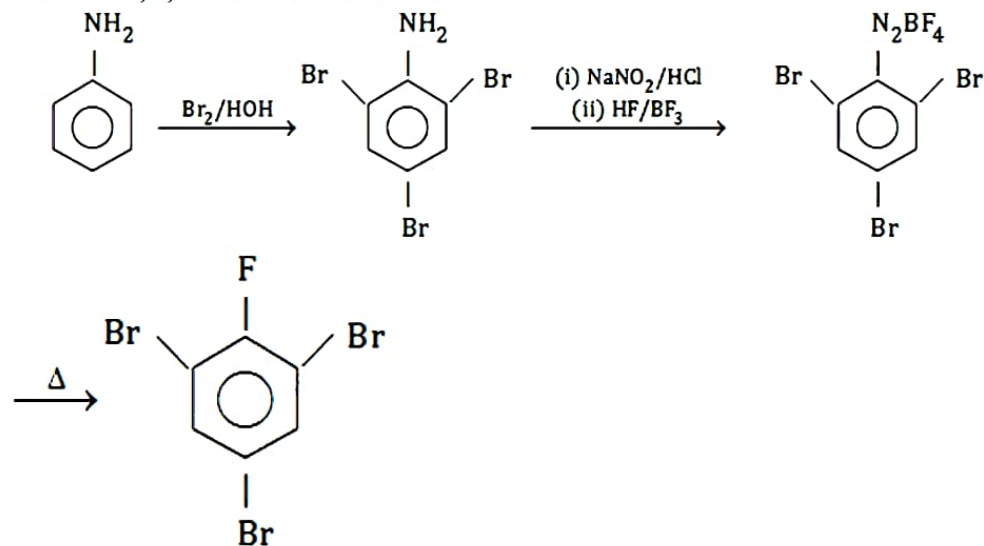
GOT IT WRONG ❌

16.30%

SOLUTION

[▶ Video Solution](#)

Here X is 2, 4, 6-tribromofluorobenzene



Match list I with list II and select the correct answer using the codes given below the lists.

List I (Pair of isomers)	List II (Type of isomerism)
(p) (I) $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ (II) $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$ }	1. Ionization
(q) (III) $[\text{PrCl}_2(\text{NH}_3)_4]\text{Br}_2$ (IV) $[\text{PrBr}_2(\text{NH}_3)_4]\text{Cl}_2$ }	2. Hydrate
(r) (v) $[\text{Co}(\text{SCN})(\text{NH}_3)_5]\text{Cl}_2$ (vi) $[\text{Co}(\text{NCS})(\text{NH}_3)_5]\text{Cl}_2$ }	3. Coordination
(s) (VII) $[\text{Cr}_2(\text{H}_2\text{O})_6]\text{Cl}_3$ (viII) $[\text{CrCl}_2(\text{H}_2\text{O})_4]\text{Cl} \cdot 2\text{H}_2\text{O}$ }	4. Geometrical
	5. Linkage isomerism

A

<i>p</i>	<i>q</i>	<i>r</i>	<i>s</i>
4	1	5	2

3.65% marked this

B

<i>p</i>	<i>q</i>	<i>r</i>	<i>s</i>
1	3	2	5

0.26% marked this

C

<i>p</i>	<i>q</i>	<i>r</i>	<i>s</i>
3	1	5	2

Correct

88.40% marked this

D

<i>p</i>	<i>q</i>	<i>r</i>	<i>s</i>
1	3	5	2

7.69% marked this

TIME TAKEN

Others: 81 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Easy

CHAPTER

Coordination Compounds

TOPIC

Isomerism of coordination compounds

DIDN'T ATTEMPT

6.35%

GOT IT RIGHT

88.40%

GOT IT WRONG

11.60%

SOLUTION

[▶ Video Solution](#)

Compounds (I) and (II) both contain complex cation and anion, so they show coordination isomerism.

Compounds (III) and (IV) are obtained by the interchange of groups outside the coordination sphere and the ligands within the coordination sphere. Therefore, they give different ions in the solution. Hence, they show ionization isomerism.

Compounds (V) and (VI) are linkage isomers, because the ambident ligands, SCN has two donor atoms, i.e., it has two possibilities of attachment (SCN and NCS) with the central metal. Hence, they show linkage isomerism.

Compounds (VII) and (VIII) are obtained by the replacement of a coordinated group such as Cl^- by water of hydration. Hence, they show hydrate isomerism.

Among the following molecules/ions, what is the correct order of increasing s-character (in percentage) in the hybrid orbitals?

(I) CO_3^{2-} (II) XeF_4 (III) I_3^- (IV) NCl_3 (V) BeCl_2

II < III < IV < I < V

A

Correct

84.39% marked this

II < IV < III < V < I

B

1.75% marked this

III < II < I < V < IV

C

0.79% marked this

II < IV < III < I < V

D

13.07% marked this

TIME TAKEN

Others: 133 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

Chemical Bonding and Molecular Structure

TOPIC

Hybridisation and VSEPR Theory

DIDN'T ATTEMPT

11.39%

GOT IT RIGHT

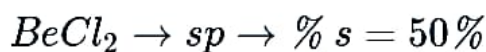
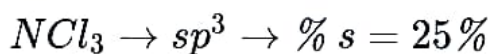
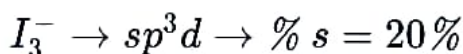
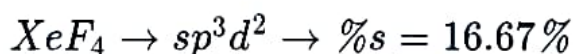
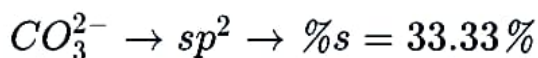
84.39%

GOT IT WRONG

15.61%

SOLUTION

Video Solution



Identify the set of acidic oxides.

- A** $\text{Na}_2\text{O}, \text{CaO}, \text{BaO}$
9.31% marked this
- B** $\text{ZnO}, \text{PbO}, \text{BeO}$
5.63% marked this
- C** $\text{CO}, \text{NO}, \text{N}_2\text{O}$
14.96% marked this
- D** $\text{Mn}_2\text{O}_7, \text{CrO}_3, \text{V}_2\text{O}_5$
Correct
70.11% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 39 sec

90 sec

DIFFICULTY

Easy

CHAPTER

Classification of Elements and
Periodicity in Properties

TOPIC

Periodic properties

DIDN'T ATTEMPT

9.03%

GOT IT RIGHT

70.11%

GOT IT WRONG

29.89%

SOLUTION

[▶ Video Solution](#)

Acidic oxides $\rightarrow \text{Mn}_2\text{O}_7, \text{CrO}_3, \text{V}_2\text{O}_5$

(due to higher oxidation state of central metal)

Basic oxide $\rightarrow \text{Na}_2\text{O}, \text{CaO}, \text{BaO}$

(due to lower oxidation state of central metal)

Alkaline and alkali metal oxide are always basic.

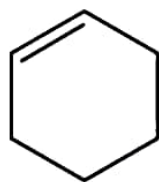
Neutral oxides $\rightarrow \text{CO}, \text{NO}, \text{N}_2\text{O}$

Amphoteric oxides $\rightarrow \text{ZnO}, \text{PbO}, \text{BeO}$

(acidic as well as basic)

Hence, option (4) is correct.

If enthalpy of hydrogenation of $\text{C}_6\text{H}_6(l)$ into $\text{C}_6\text{H}_{12}(l)$ is -205 kJ and resonance energy of $\text{C}_6\text{H}_6(l)$ is -152 kJ mol^{-1} then enthalpy of hydrogenation of



(l) to $\text{C}_6\text{H}_{12}(l)$ is?

[Assume enthalpies of vapourisation of all liquids involved are equal].

A -535.5 kJ/mol
0.59% marked this

B -238 kJ/mol
3.82% marked this

C -357 kJ/mol
20.27% marked this

D -119 kJ/mol
Correct
75.32% marked this

EVALUATION

TIME TAKEN

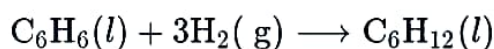
TIME ALLOTTED

Others: 116 sec

120 sec

DIFFICULTY
MediumCHAPTER
Thermodynamics (C)TOPIC
Laws of Thermochemistry and
Enthalpy ChangeDIDN'T ATTEMPT
47.13%GOT IT RIGHT
75.32%GOT IT WRONG
24.68%

SOLUTION

[▶ Video Solution](#)

$$\Delta H_{\text{hyd}}^\circ(\text{C}_6\text{H}_6)_l = -205 \text{ kJ}$$

Benzene on reactant side so add resonance energy of benzene with sign.

Heat evolved for hydrogenation of three double bonds

$$= -205 - 152 = -357 \text{ kJ mol}^{-1}$$

$$\Delta H_{\text{hyd}}^\circ = \frac{-357}{3} = -119 \text{ kJ/mole}$$

Assertion: La ($Z = 57$) belongs to the d-block and is an element of group 3 of the periodic table.

Reason: All Lanthanoids belong to the group 3 of the periodic table.

Both the assertion and reason statements are true and the reason is the correct explanation of the assertion

A

37.08% marked this

Both assertion and reason statements are true but the reason is not the correct explanation of the assertion.

B

Correct

47.72% marked this

The assertion statement is true but the reason statement is false

C

9.26% marked this

Both the assertion and reason statements are false

D

5.94% marked this

EVALUATION

TIME TAKEN

Others: 36 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Easy

CHAPTER

d and f Block Elements

TOPIC

F-Block elements

DIDN'T ATTEMPT 🕒

7.82%

GOT IT RIGHT ✅

47.72%

GOT IT WRONG ❌

52.28%

SOLUTION

[▶ Video Solution](#)

All lanthanoids belonging to the 3rd group of the periodic table do not justify why Lanthanum is called a d-block metal. It is called a d-block metal because the differentiating electron enters the d-orbit.

Hence, both the assertion and reason statements are true but the reason is not the correct explanation of the assertion.

For a complex reaction



$$E_{a1} = 200 \text{ kJ/mol}, E_{a2} = 90 \text{ kJ/mol}, E_{a3} = 80 \text{ kJ/mol}$$

Overall rate constant k is related to individual rate constant by the equation

$$k = \left(\frac{k_1 k_2}{k_3} \right)^{2/3}$$

Activation energy (kJ/mol) for the overall reaction is

(mark answer to nearest integer)

A

240

1.56% marked this

B

160

0.83% marked this

C

210

11.40% marked this

D

140

Correct

86.21% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 116 sec

180 sec

DIFFICULTY

CHAPTER

TOPIC

Hard

Chemical Kinetics

Arrhenius theory

DIDN'T ATTEMPT

GOT IT RIGHT

GOT IT WRONG

29.43%

86.21%

13.79%

SOLUTION

[▶ Video Solution](#)

Problem includes concept of Arrhenius equation for different rate constant of a common reaction. To solve this problem student is advised to write the Arrhenius equation for every rate constant. Then, put the value in

$$k = \left(\frac{k_1 k_2}{k_3} \right)^{2/3} \text{ to get the value of } E_a$$

$$\text{Putting } k = A e^{-E_a/RT}$$

$$k_1 = A_1 e^{-\frac{E_{a1}}{RT}}, k_2 = A_2 e^{-\frac{E_{a2}}{RT}}$$

$$k_3 = A_3 e^{-\frac{E_{a3}}{RT}}$$

$$k = \left(\frac{A_1 e^{-\frac{E_{a1}}{RT}} \cdot A_2 e^{-\frac{E_{a2}}{RT}}}{A_3 e^{-\frac{E_{a3}}{RT}}} \right)^{2/3} = A e^{-E_a/RT}$$

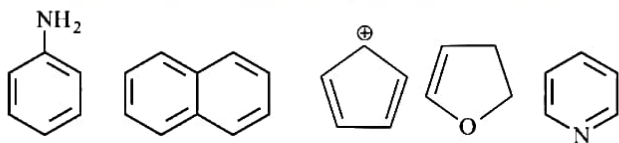
Comparing coefficients

$$e^{-\frac{1}{RT} \left\{ \frac{2}{3} (E_{a1} + E_{a2}) - \frac{2}{3} E_{a3} \right\}} = e^{-E_a/RT}$$

$$E_a = \frac{2}{3} [E_{a1} + E_{a2} - E_{a3}]$$

$$E_a = \frac{2}{3} [200 + 90 - 80] = \frac{2}{3} [210] \\ = 140 \text{ kJ/mol}$$

Total number of aromatic compounds from below is



- A** 1
0.86% marked this
- B** 4
7.14% marked this
- C** 2
8.33% marked this
- D** 3
Correct
83.67% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 46 sec

120 sec

DIFFICULTY

Medium

CHAPTER

General Organic Chemistry

TOPIC

Conjugation and Aromaticity

DIDN'T ATTEMPT

4.47%

GOT IT RIGHT

83.67%

GOT IT WRONG

16.33%

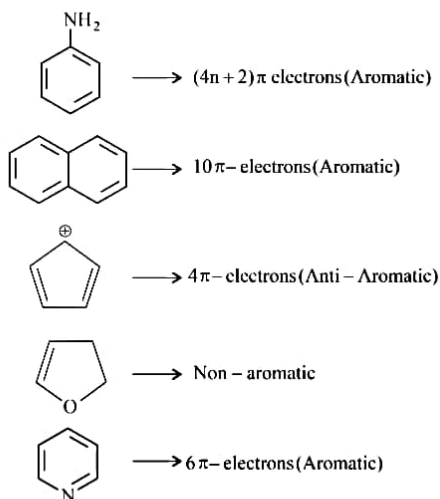
SOLUTION

[▶ Video Solution](#)

According to the Huckle's rule,

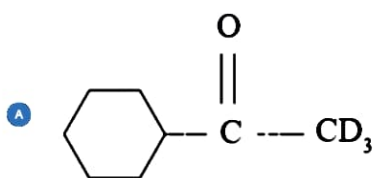
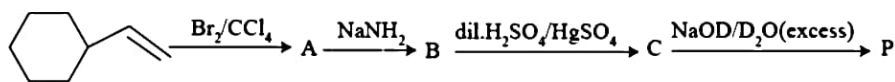
The compound is aromatic which is planar and has the $(4n + 2)\pi$ electrons. The compound is aromatic in nature because it is a planar compound with $(4n + 2)\pi$ electrons. Aniline, naphthalene and pyridine are aromatic compounds.

Compounds :-

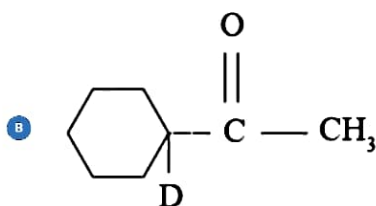


Hence, the three compounds are aromatic in nature.

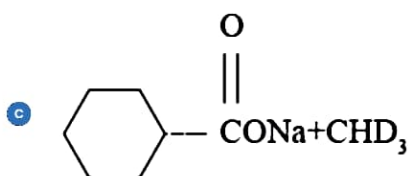
Final product (p) in the sequence of reaction is



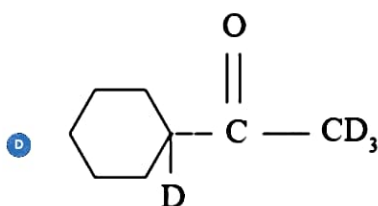
22.07% marked this



7.99% marked this



14.79% marked this



Correct
55.15% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 74 sec

210 sec

DIFFICULTY

Hard

CHAPTER

Hydrocarbons

TOPIC

Reactions of alkenes

DIDN'T ATTEMPT

42.85%

GOT IT RIGHT

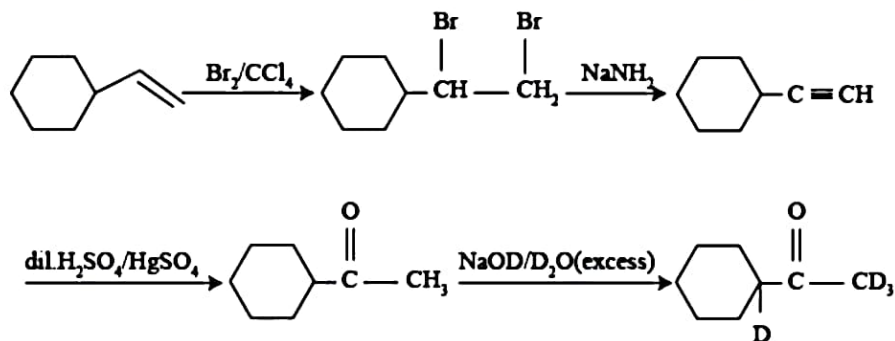
55.15%

GOT IT WRONG

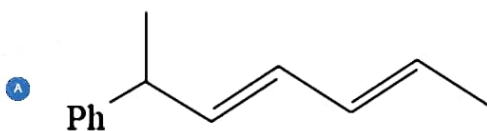
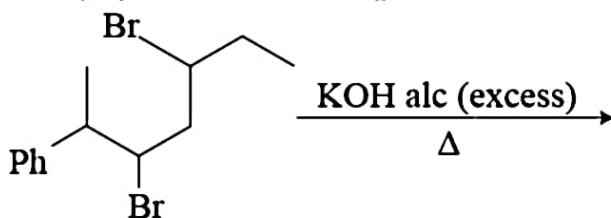
44.85%

SOLUTION

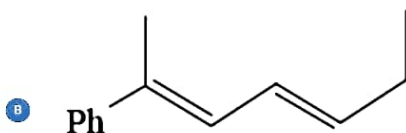
[▶ Video Solution](#)



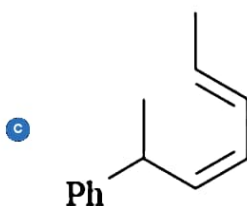
The major product of the following reaction is



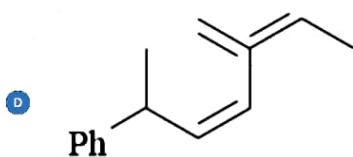
7.83% marked this



87.49% marked this



4.15% marked this



0.54% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 74 sec

120 sec

DIFFICULTY
Medium

CHAPTER
Haloalkanes and Haloarenes

TOPIC
Preparation and Properties of Monohalides

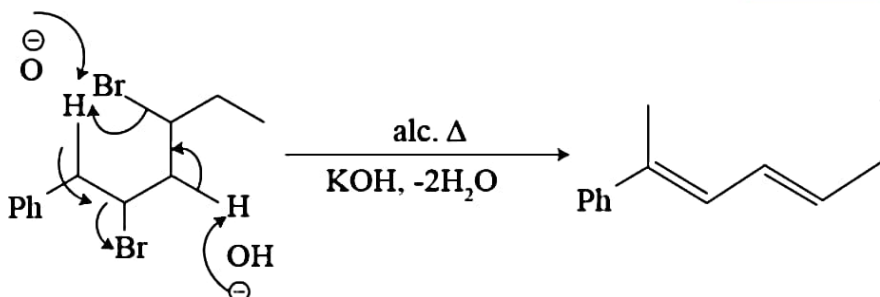
DIDN'T ATTEMPT
12.94%

GOT IT RIGHT
87.49%

GOT IT WRONG
12.51%

SOLUTION

[▶ Video Solution](#)



This product is highly stable conjugate di-ene system with benzene ring also so it is formed here as major product.

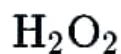
Select the compound which cannot act as an oxidising agent and reducing agent?

A



6.36% marked this

B



9.30% marked this

C



71.24% marked this

D



13.10% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 48 sec

90 sec

DIFFICULTY

CHAPTER

TOPIC

Easy

Redox Reactions

Oxidation and Reduction Reactions

DIDN'T ATTEMPT



GOT IT RIGHT



GOT IT WRONG



17.22%

71.24%

28.76%

SOLUTION

[▶ Video Solution](#)

In HClO_4 , Cl is at maximum oxidation state so can't act as reducing agent.

The coordination number of Ni^{2+} is 4 .

$\text{NiCl}_2 + \text{KCN (excess)} \rightarrow A$ (cyano complex)

$\text{NiCl}_2 + \text{conc. HCl (excess)} \rightarrow B$ (chloro complex)

The IUPAC name of A and B are

Potassium tetracyanonickelate (II), potassium tetrachloronickelate (II)

A

Correct

81.24% marked this

Tetracyanopotassiumnickelate (II), tetrachloropotassiumnickelate (II)

B

1.93% marked this

Tetracyanonickel (II), tetrachloronickel (II)

C

10.33% marked this

Potassium tetracyanonickel (II), potassium tetrachloronickel (II)

D

6.50% marked this

EVALUATION

TIME TAKEN

Others: 97 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Hard

CHAPTER

Coordination Compounds

TOPIC

Werner's theory and Valence Bond Theory

DIDN'T ATTEMPT

23.06%

GOT IT RIGHT

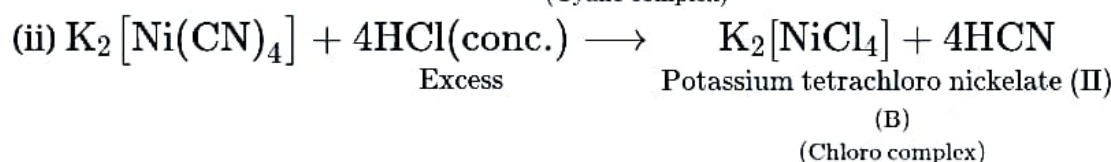
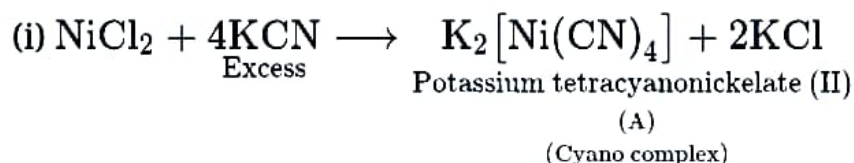
81.24%

GOT IT WRONG

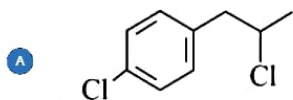
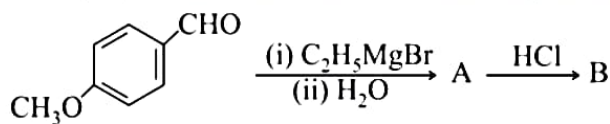
18.76%

SOLUTION

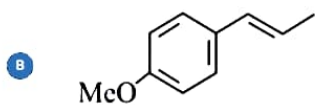
Video Solution



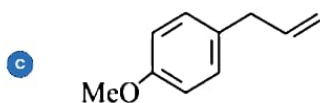
The major product B formed in the following reaction sequence is:



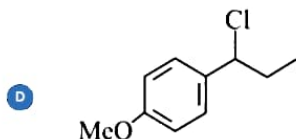
1.91% marked this



17.19% marked this



0.62% marked this



80.29% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

150 sec

Others: 84 sec

DIFFICULTY

Medium

CHAPTER

Aldehydes and Ketones

TOPIC

Method of preparation for both aldehydes and ketones

DIDN'T ATTEMPT

16.14%

GOT IT RIGHT

80.29%

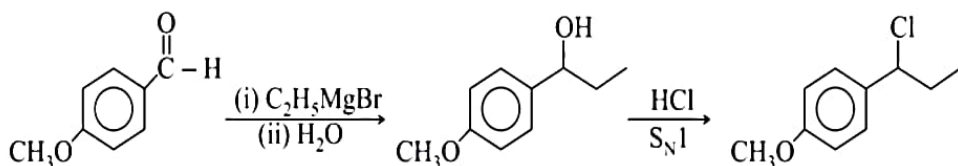
GOT IT WRONG

19.71%

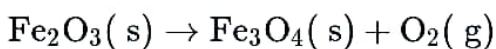
SOLUTION

[▶ Video Solution](#)

Initially aldehyde reacts with Grignard reagent to form a secondary alcohol. Which undergo nucleophilic substitution reaction.



1.5gm mixture of SiO_2 and Fe_2O_3 on very strong heating leave a residue weighting 1.46gm. The reaction responsible for loss of weight is :



What is the percentage by mass of Fe in original sample :-

Your Answer

Correct Answer

56

TIME TAKEN

Others: 44 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Medium

CHAPTER

Some Basic Concepts of Chemistry

TOPIC

Quantitative measures in chemical equations

DIDN'T ATTEMPT 

73.36%

GOT IT RIGHT 

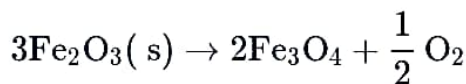
13.95%

GOT IT WRONG 

86.05%

SOLUTION

[▶ Video Solution](#)



3 mol

0.5 mol

$3 \times 160 \text{ gm}$

$\frac{1}{2} \times 32 \text{ gm}$

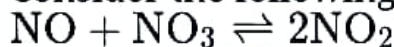
loss of 16 gm O_2 by 480 gm Fe_2O_3

loss of 0.04 gm $\text{O}_2 \rightarrow 0.04 \times \frac{480}{16} \text{ Fe}_2\text{O}_3$

$= 1.2 \text{ gm Fe}_2\text{O}_3$

% by mass of Fe $= \frac{0.84}{1.5} \times 100 = 56\%$

Consider the following reversible reaction:



If 1.0 mol of NO is mixed with 3.0 mol of NO_3 , 'x' mole of NO_2 is produced at equilibrium. If 2.0 mol of NO is added further, 'x' mol of NO_2 is further produced. What is the value of equilibrium constant?

Your Answer

Correct Answer

4

TIME TAKEN

Others: 175 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Medium

CHAPTER

Chemical Equilibrium

TOPIC

Law of Mass Action, Equilibrium Constant (K_c and K_p) and its Application

DIDN'T ATTEMPT

66.70%

GOT IT RIGHT

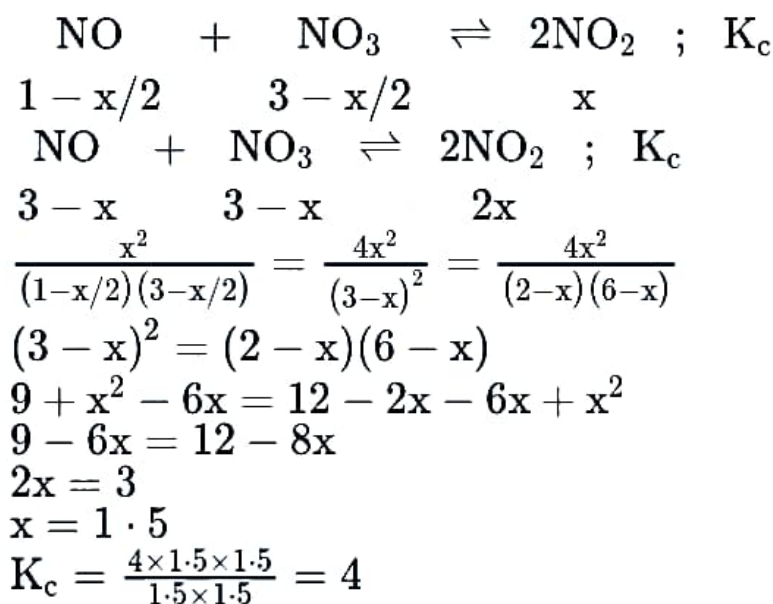
61.98%

GOT IT WRONG

38.02%

SOLUTION

▶ Video Solution



A solution containing 8.0 g of nicotine in 92 g of water freezes 0.925 degrees below the normal freezing point of water. If the molal freezing point depression constant, $K_f = 1.85^\circ\text{C mol}^{-1}$, then the molar mass of nicotine is: (mark answer to nearest integer)

Your Answer

Correct Answer

174

TIME TAKEN

Others: 121 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Easy

CHAPTER

Solutions

TOPIC

Colligative Properties and Abnormal Molecular Masses

DIDN'T ATTEMPT 

21.97%

GOT IT RIGHT 

53.84%

GOT IT WRONG 

46.16%

SOLUTION

[▶ Video Solution](#)

When a non-volatile solute is added in a pure solvent, the freezing point of the solution decreases. This decrease in the freezing point is known as depression in freezing point.

Here,

Weight of solute, $W_B = 8\text{ g}$ Weight of solvent, $W_A = 92\text{ g} = 0.092\text{ kg}$ Molal freezing point depression constant, $K_f = 1.85^\circ\text{C mol}^{-1}$

Depression in freezing point,

$$\Delta T_f = K_f m \Rightarrow \Delta T_f = K_f \times \frac{W_B}{M_B} \times \frac{1}{W_A} \Rightarrow M_B = \frac{1.85 \times 8}{0.925 \times 0.092} = 173.9$$

Hence, the molar mass of nicotine = 174 g mol^{-1} .

For the following electrochemical cell at 298 K,
 $\text{Pt(s)}|\text{H}_2(\text{g}, 1 \text{ bar})|\text{H}^+(\text{aq}, 1\text{M})||\text{M}^{4+}(\text{aq}), \text{M}^{2+}(\text{aq})|\text{Pt(s)}$

$$E_{\text{cell}} = 0.092\text{V} \text{ when } \frac{[\text{M}^{2+}(\text{aq})]}{[\text{M}^{4+}(\text{aq})]} = 10^x$$

$$\text{Given: } E_{\text{M}^{4+}|\text{M}^{2+}}^0 = 0.151\text{V}; 2.303 \frac{RT}{F} = 0.059 \text{ V}$$

The value of x is (nearest integer).....

Your Answer

Correct Answer

2

TIME TAKEN

Others: 116 sec

TIME ALLOTTED

240 sec

DIFFICULTY

Hard

CHAPTER

Electrochemistry

TOPIC

Cells and Electrode Potential, Nernst Equation

DIDN'T ATTEMPT

26.34%

GOT IT RIGHT

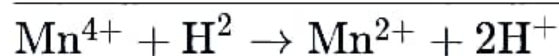
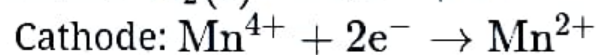
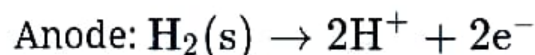
84.93%

GOT IT WRONG

15.07%

SOLUTION

▶ Video Solution



$$E = E^\circ - \frac{0.059}{2} \log_{10} \left(\frac{[\text{Mn}^{2+}][\text{H}^+]^2}{[\text{Mn}^{4+}]\text{P}_{\text{H}_2}} \right)$$

$$0.092 = 0.151 - \frac{0.059}{2} \log_{10} (10^x)$$

$$0.092 = 0.151 - \frac{0.059}{2} x$$

$$x = 2$$

There are some isomeric amines containing only one N atom. Each one of which forms a chloride salt on treatment with HCl containing 32.42% chlorine. None of them decolorizes Br_2 water. How many of them does not give carbyl amine test?

Your Answer

Correct Answer

4

TIME TAKEN

Others: 75 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Hard

CHAPTER

Amines

TOPIC

Preparation and Properties of Amines

DIDN'T ATTEMPT 

80.96%

GOT IT RIGHT 

64.43%

GOT IT WRONG 

35.57%

SOLUTION

[▶ Video Solution](#)

Let us assume molwt of amine hydrochloride = M

$$M \times \frac{32.4}{100} = 35.5$$

$$\therefore M = \frac{35.5 \times 100}{32.4}$$

$$= 109.5$$

$\therefore$ mol mass of the amines

= mol mass of amine hydrochloride – mol mass of HCl

$$= 109.5 - 36.5$$

$$= 73.$$

Let the formula be RNH_2 or $\text{C}_n\text{H}_{2n+1}\text{NH}_2$

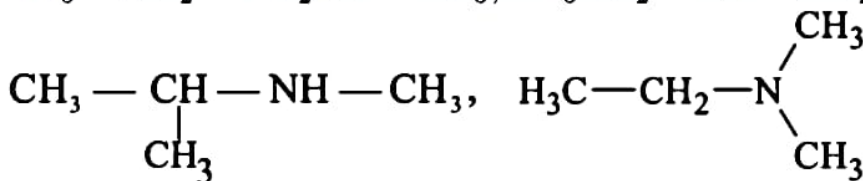
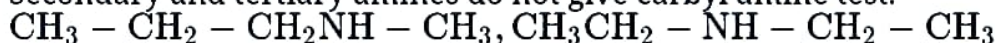
$$\therefore 12 \times n + 1 \times (2n + 1) + 16 = 73$$

$$14n = 56$$

$$n = 4$$

Mol formula of amines = $\text{C}_4\text{H}_9\text{NH}_2$

Secondary and tertiary amines do not give carbyl amine test.



Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = -x^3 - 3x^2 - 6x + 1$.
 Number of integers in the solution set of x satisfying the inequality
 $f(f(x^3 + f(x))) \geq f(f(-f(x) - x^3))$ is?

3

A

Correct

88.86% marked this

4

B

5.19% marked this

5

C

2.59% marked this

6

D

3.36% marked this

TIME TAKEN

Others: 260 sec

TIME ALLOTTED

210 sec

DIFFICULTY

Hard

CHAPTER

Application of Derivatives

TOPIC

Monotonicity

DIDN'T ATTEMPT



61.82%

GOT IT RIGHT



88.86%

GOT IT WRONG



11.14%

SOLUTION

Video Solution

$$f'(x) < 0$$

$$\therefore f(x^3 + f(x)) \leq f(-f(x) - x^3)$$

$$\Rightarrow x^3 + f(x) \geq -f(x) - x^3$$

$$\Rightarrow f(x) + x^3 \geq 0$$

$$\Rightarrow 3x^2 + 6x - 1 \leq 0$$

$$\text{As } x \in \mathbb{Z}, x \in \{-2, -1, 0\}$$

The system of simultaneous equations $kx + 2y - z = 1$, $(k - 1)y - 2z = 2$ and $(k + 2)z = 3$ has a unique solution, if k equals

A -2

4.22% marked this

B -1

Correct

85.43% marked this

C 0

4.75% marked this

D 1

5.60% marked this

TIME TAKEN

Others: 120 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Easy

CHAPTER

Determinants

TOPIC

System of Linear Equations

DIDN'T ATTEMPT 

17.96%

GOT IT RIGHT 

85.43%

GOT IT WRONG 

14.57%

SOLUTION

 Video Solution

The matrix form of the given system of equation is:

$$\begin{bmatrix} k & 2 & -1 \\ 0 & k-1 & -2 \\ 0 & 0 & k+2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Therefore, the coefficient matrix will have a solution iff it's determinant $\neq 0$

i.e. $(k - 1)(k + 2) \neq 0$

i.e., $k \neq 0, 1, -2$

Thus the given system of equation will have a solution only at $k = -1$.

If $f(x) = \cos x - \int_0^x (x-t)f(t)dt$, then $f''(x) + f(x)$ is equal to

— $\cos x$

A

Correct

54.69% marked this

— $\sin x$

B

2.45% marked this

$\int_0^x (x-t)f(t)dt$

C

30.58% marked this

0

D

12.27% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 89 sec

210 sec

DIFFICULTY

Medium

CHAPTER

Definite Integration

TOPIC

Leibnitz Rule of Differentiation

DIDN'T ATTEMPT



42.37%

GOT IT RIGHT



54.69%

GOT IT WRONG



45.31%

SOLUTION

▶ Video Solution

$$f(x) = \cos x - \int_0^x (x-t)f(t)dt$$

$$\Rightarrow f(x) = \cos x - x \int_0^x f(t)dt + \int_0^x tf(t)dt$$

(differentiating w.r.t. x. by Leibnitz's Rule)

$$\Rightarrow f'(x) = -\sin x - x f(x) - \int_0^x f(t)dt + x f(x)$$

$$\Rightarrow f'(x) = -\sin x - \int_0^x f(t)dt$$

(again differentiating w.r.t.x)

$$\Rightarrow f''(x) = -\cos x - f(x)$$

$$\Rightarrow f''(x) + f(x) = -\cos x$$

If z is a complex number such that $z + i|z| = i\bar{z} + 1$, then $|z|$ is -

A 4
0.59% marked this

B 2
1.95% marked this

C 1
Correct
95.20% marked this

D 0
2.27% marked this

TIME TAKEN

Others: 130 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Medium

CHAPTER

Complex Number

TOPIC

Conjugate, modulus and argument

DIDN'T ATTEMPT ☹️

38.85%

GOT IT RIGHT ✅

95.20%

GOT IT WRONG ❌

4.80%

SOLUTION

▶ Video Solution

Put $z = x + iy$

$$(x + iy) + i(|z|) = i(x - iy) + 1$$

$$\Rightarrow x = y + 1 \& y + |z| = x$$

$$\Rightarrow x - y = 1 \& x - y = |z|$$

$$\Rightarrow |z| = 1$$

If $\alpha_1, \beta_1, \gamma_1, \delta_1$ are the roots of the equation $ax^4 + bx^3 + cx^2 + dx + e = 0$ and $\alpha_2, \beta_2, \gamma_2, \delta_2$ are the roots of the equation $ex^4 + dx^3 + cx^2 + bx + a = 0$ such that $0 < \alpha_1 < \beta_1 < \gamma_1 < \delta_1, 0 < \alpha_2 < \beta_2 < \gamma_2 < \delta_2$ $\alpha_1 - \delta_2 = 2 = \beta_1 - \gamma_2; \gamma_1 - \beta_2 = \delta_1 - \alpha_2 = 4$, then $a + b + c + d + e =$

A 10
14.29% marked this

B 12
23.29% marked this

C 6
10.87% marked this

D 8
Correct
51.55% marked this

TIME TAKEN

Others: 137 sec

TIME ALLOTTED

210 sec

DIFFICULTY
HardCHAPTER
Quadratic EquationTOPIC
N degree equationDIDN'T ATTEMPT
78.41%GOT IT RIGHT
51.55%GOT IT WRONG
48.45%

SOLUTION

[▶ Video Solution](#)

Given,

$\alpha_1, \beta_1, \gamma_1, \delta_1$ are the roots of equation $ax^4 + bx^3 + cx^2 + dx + e = 0$ and $\alpha_2, \beta_2, \gamma_2, \delta_2$ are the roots of equation $ex^4 + dx^3 + cx^2 + bx + a = 0$

 $\therefore$ Clearly,

$\alpha_1, \beta_1, \gamma_1, \delta_1$ are the reciprocal of roots of $ax^4 + bx^3 + cx^2 + dx + e = 0$

Also, $\delta_1 > \gamma_1 > \beta_1 > \alpha_1 > 0$ and $\delta_2 > \gamma_2 > \beta_2 > \alpha_2 > 0$ $\therefore \delta_1$ and α_2 are reciprocalSimilarly γ_1 and β_2 , γ_2 and β_1 and α_1 and δ_2 are reciprocal $\therefore \alpha_1 \delta_2 = 1, \beta_1 \gamma_2 = 1, \beta_2 \gamma_1 = 1, \alpha_2 \delta_1 = 1$

$$\alpha_1 - \delta_2 = 2$$

$$\therefore \alpha_1 - \frac{1}{\alpha_1} = 2 \Rightarrow \alpha_1^2 - 2\alpha_1 - 1 = 0$$

$$\text{and } \delta_1 - \alpha_2 = 4$$

$$\delta_1 - \frac{1}{\delta_1} = 4 = \delta_1^2 - 4\delta_1 - 1 = 0$$

 $\therefore \alpha_1$ and δ_1 are roots of

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

$$\therefore (x^2 - 2x - 1)(x^2 - 4x - 1) = ax^4 + bx^3 + cx^2 + dx + e$$

Put $x = 1$,

$$(1 - 2 - 1)(1 - 4 - 1) = a + b + c + d + e$$

$$8 = a + b + c + d + e$$

If the area bounded by $f(x) = \tan^3 x + \tan x$ from $x = 0$ to $x = \frac{\pi}{4}$ is k square units, then the maximum value of $g(x) = k \sin x$ is $\left(\forall x \in \left[0, \frac{\pi}{4}\right]\right)$

A

2

0.86% marked this

B

 $\frac{1}{2\sqrt{2}}$

Correct

98.27% marked this

C

4

0.66% marked this

D

8

0.21% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 142 sec

210 sec

DIFFICULTY

Hard

CHAPTER

Area Under Curves

TOPIC

Area bounded by two curves

DIDN'T ATTEMPT

40.71%

GOT IT RIGHT

98.27%

GOT IT WRONG

1.73%

SOLUTION

▶ Video Solution

The required area is

$$A = \int_0^{\frac{\pi}{4}} (\tan^3 x + \tan x) dx$$

$$= \int_0^{\frac{\pi}{4}} \tan x (\tan^2 x + 1) dx$$

$$= \int_0^{\frac{\pi}{4}} \tan x \cdot \sec^2 x dx$$

$$\text{Let } \tan x = t$$

$$\Rightarrow \sec^2 x dx = dt$$

$$\Rightarrow I = \int_0^1 t dt = \left[\frac{t^2}{2} \right]_0^1$$

$$= \frac{1}{2}$$

$$\therefore g(x) = \frac{1}{2} \sin x \leq \frac{1}{2\sqrt{2}}$$

Let the function defined as $f(x) = \begin{cases} \frac{\sin[x^2]\pi}{x^2-3x-18} + ax^3 + b, & 0 \leq x \leq 1 \\ 2\cos\pi x + \tan^{-1}x, & 1 < x \leq 2 \end{cases}$ is a differentiable function in the interval $(0, 2)$, then the value of $[a + b]$ equals, ($[\cdot]$ represents greatest integer function)

A 0

3.02% marked this

B -1

9.36% marked this

C -2

Correct
84.87% marked this

D 1

2.75% marked this

EVALUATION

TIME TAKEN

You:
Others: 142 sec

TIME ALLOTTED

150 sec

DIFFICULTY
Medium

CHAPTER
Continuity and Differentiability

TOPIC
Differentiability

DIDN'T ATTEMPT
42.98%

GOT IT RIGHT
84.87%

GOT IT WRONG
15.13%

SOLUTION

▶ Video Solution

Since the value of $[x]$ is always an integer and $\sin n\pi = 0$, $n \in I$, therefore, $\sin[x^3]\pi = 0$.

$$f(x) = \begin{cases} ax^3 + b, & 0 \leq x \leq 1 \\ 2\cos\pi x + \tan^{-1}x, & 1 < x \leq 2 \end{cases}$$

The given function is differentiable in $(0, 2)$.

$\therefore f(x)$ is continuous at $x = 1$

$$\therefore \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$$

$$\therefore f(1) = \text{RHL} \Rightarrow a + b = 2\cos\pi + \tan^{-1}(1)$$

$$\Rightarrow a + b = -2 + \frac{\pi}{4}$$

$$\Rightarrow [a + b] = -2$$

4 cards are drawn from a well shuffled pack of 52 cards without replacement. If the fourth card is a king, then the chance that the first 3 cards are also king is $\frac{1}{20820+a}$, then a is equal to

- ☒ **A** 4
 8.37% marked this
- ☐ **B** 1
 7.39% marked this
- ☐ **C** 2
 4.88% marked this
- ☐ **D** 5
 79.36% marked this

EVALUATION

TIME TAKEN

 You:
 Others: 206 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Easy

CHAPTER

Probability

TOPIC

 Conditional Probability and
 Multiplication Theorem

DIDN'T ATTEMPT

72.06%

GOT IT RIGHT

79.36%

GOT IT WRONG

20.64%

SOLUTION

[▶ Video Solution](#)

Let A : First 3 cards are king

B : Fourth card is a king

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Let k : drawing of a king

B can occur in four types

Type 1 : $\bar{k}\bar{k}\bar{k}k \rightarrow 1$

Type 2 : $k\bar{k}\bar{k}k \rightarrow 3$

Type 3 : $k\bar{k}kk \rightarrow 3$

Type 4 : $kkkk \rightarrow 1$

$$\begin{aligned}
 P(A/B) &= \frac{\frac{1}{{}^{52}C_4}}{\frac{{}^{48}C_3 \times {}^4C_1}{4 \times {}^{52}C_4} + \frac{{}^{48}C_2 \times {}^4C_2 \times 3}{{}^{52}C_4 \times 6} + \frac{{}^{48}C_1 \times {}^4C_3 \times 3}{{}^{52}C_4 \times 4} + \frac{1}{{}^{52}C_4}} \\
 &= \frac{1}{20825} \\
 \therefore a &= 5
 \end{aligned}$$

There are 3 identical black balls, 4 identical white balls and 2 identical red balls. The number of ways they can be arranged in a row so that at least one ball separates the balls of the same colour is

A 1210
2.67% marked this

B 1212
3.47% marked this

C 1244
3.92% marked this

D 1254
Correct
89.94% marked this

TIME TAKEN

You:
Others: 128 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Easy

CHAPTER

Permutation Combination

TOPIC

Arrangement under Constraint

DIDN'T ATTEMPT

63.63%

GOT IT RIGHT

89.94%

GOT IT WRONG

10.06%

SOLUTION

Video Solution

The number of ways they can be arranged in a row is $\frac{9!}{2!3!4!} = 1260$

The three colour blocks of balls can be arranged in $3! = 6$ ways.

∴ The desired number of ways is $1260 - 6 = 1254$.

If the eccentricity of the hyperbola $x^2 - y^2 \sec^2 \theta = 5$ is $\sqrt{3}$ times the eccentricity of the ellipse $x^2 \sec^2 \theta + y^2 = 25$, then smallest positive value of θ is $\frac{\pi}{P}$, where twice the value of 'P' is

8

A

Correct

89.94% marked this

12

B

6.18% marked this

16

C

250% marked this

24

D

138% marked this

TIME TAKEN

Others: 193 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Medium

CHAPTER

Hyperbola

TOPIC

Equation of hyperbola

DIDN'T ATTEMPT

51.15%

GOT IT RIGHT

89.94%

GOT IT WRONG

10.06%

SOLUTION

Video Solution

Eccentricity of the hyperbola $x^2 - y^2 \sec^2 \theta = 5$ is

$$e_1 = \sqrt{\frac{1 + \sec^2 \theta}{\sec^2 \theta}} = \sqrt{1 + \cos^2 \theta}$$

Eccentricity of the ellipse $x^2 \sec^2 \theta + y^2 = 25$ is

$$e_2 = \sqrt{\frac{\sec^2 \theta - 1}{\sec^2 \theta}} = |\sin \theta| \text{ Given } e_1 = \sqrt{3} e_2$$

$$\Rightarrow 1 + \cos^2 \theta = 3 \sin^2 \theta \Rightarrow \cos \theta = \pm \frac{1}{\sqrt{2}}$$

$\therefore$ Least positive value of θ is $\frac{\pi}{4} \therefore P = 4 \Rightarrow 2P = 8$

An equilateral triangle's sides increase at the rate of 2 cm/sec . If the area of its incircle increases at a rate of $k \text{ cm}^2/\text{sec}$ (when the length of the side is $\frac{6}{\pi} \text{ cm}$), then the value of k is

A 1
2.88% marked this

B 4
6.72% marked this

C 2
Correct
85.14% marked this

D 3
5.26% marked this

EVALUATION

TIME TAKEN

Others: 145 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Easy

CHAPTER

Application of Derivatives

TOPIC

Rate Measure Error and Approximation

DIDN'T ATTEMPT

35.26%

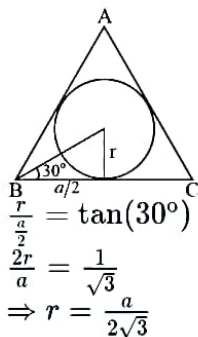
GOT IT RIGHT

85.14%

GOT IT WRONG

14.86%

SOLUTION

[▶ Video Solution](#)

$$\therefore \text{Area of incircle, } A = \pi r^2 = \frac{\pi a^2}{12}$$

$$\begin{aligned} \therefore \frac{dA}{dt} &= \frac{2\pi a}{12} \frac{da}{dt} \\ &= \frac{\pi}{6} \cdot \frac{6}{\pi} \cdot 2 \\ &= 2 \text{ cm}^2/\text{sec}. \end{aligned}$$

If $\sum_{i=1}^5 (x_i - 6) = 5$ and $\sum_{i=1}^5 (x_i - 6)^2 = 25$, then the standard deviation of observations $3x_1 + 2$, $3x_2 + 2$, $3x_3 + 2$, $3x_4 + 2$ and $3x_5 + 2$ is equal to

A 4
4.84% marked this

B 6
Correct
88.83% marked this

C 3
2.93% marked this

D 8
3.41% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

180 sec

Others: 208 sec

DIFFICULTY

Medium

CHAPTER

Statistics

TOPIC

Measures of Dispersion

DIDN'T ATTEMPT ☹️

53.50%

GOT IT RIGHT ✅

88.83%

GOT IT WRONG ❌

11.17%

SOLUTION

▶ Video Solution

$$\text{Variance of } (3x_i + 2) = 9\text{var}(x_i)$$

$$= 9 \left(\frac{\sum x_i^2}{5} - \left(\frac{\sum x_i}{5} \right)^2 \right)$$

$$= 9(5 - 1)$$

$$= 36$$

$$\Rightarrow \text{S.D} = 6$$

Consider the matrix $A = \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix}$ and B be a square matrix of order 2 such that $BB^T = B^TB = I$. Let $C = BAB^T$ and $D = [d_{ij}]_{2 \times 2} = B^T C^6 B$, then d_{11} is equal to

A 25
0.11% marked this

B 27
Correct
96.19% marked this

C 30
0.52% marked this

D 81
3.18% marked this

TIME TAKEN

Others: 188 sec

TIME ALLOTTED

210 sec

DIFFICULTY
HardCHAPTER
MatricesTOPIC
Inverse of a MatrixDIDN'T ATTEMPT 
32.95%GOT IT RIGHT 
96.19%GOT IT WRONG 
3.81%

SOLUTION

[▶ Video Solution](#)

$$D = B^T (BAB^T) (BAB^T) (BAB^T) (BAB^T) (BAB^T) (BAB^T) B$$

$$D = A^6$$

$$\text{Now, } A^2 = \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2\sqrt{3} - 2 \\ 0 & 1 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 3 & -2\sqrt{3} - 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3\sqrt{3} & -2\sqrt{3} - 8 \\ 0 & 1 \end{bmatrix}$$

$$A^6 = A^3 \cdot A^3 = \begin{bmatrix} 3\sqrt{3} & -2\sqrt{3} - 8 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3\sqrt{3} & -2\sqrt{3} - 8 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 27 & -26 - 26\sqrt{3} \\ 0 & 1 \end{bmatrix} = D$$

$$\Rightarrow d_{11} = 27$$

$A(1, 0)$ and $B(0, 1)$ and two fixed points on the circle $x^2 + y^2 = 1$. C is a variable point on this circle. As C moves, the locus of the orthocentre of the triangle ABC is

A

$$x^2 + y^2 - 2x - 2y + 1 = 0$$

Correct

73.41% marked this

B

$$x^2 + y^2 - x - y = 0$$

21.01% marked this

C

$$x^2 + y^2 = 4$$

1.18% marked this

D

$$x^2 + y^2 + 2x - 2y + 1 = 0$$

4.40% marked this

TIME TAKEN

Others: 153 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Medium

CHAPTER

Circle

TOPIC

Locus

DIDN'T ATTEMPT

55.10%

GOT IT RIGHT

73.41%

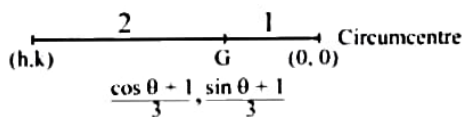
GOT IT WRONG

26.59%

SOLUTION

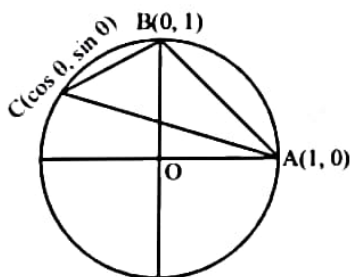
▶ Video Solution

Let $C(\cos \theta, \sin \theta)$; $H(h, k)$ is the orthocentre of the $\triangle ABC$



$$h = 1 + \cos \theta$$

$$k = 1 + \sin \theta$$



$$(x - 1)^2 + (y - 1)^2 = 1$$

$$x^2 + y^2 - 2x - 2y + 1 = 0$$

If A and B are square matrices of order 3×3 , where $|A| = 2$ and $|B| = 3$, then $\left| (A^{-1}) \operatorname{adj}(B^{-1}) \cdot \operatorname{adj}(2A^{-1}) \right| =$

A $\frac{8}{9}$

Correct

96.86% marked this

B $\frac{7}{9}$

0.59% marked this

C $\frac{2}{3}$

238% marked this

D $\frac{5}{3}$

0.17% marked this

TIME TAKEN

Others: 141 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Easy

CHAPTER

Matrices

TOPIC

Adjoint and its Properties

DIDN'T ATTEMPT

28.98%

GOT IT RIGHT

96.86%

GOT IT WRONG

3.14%

SOLUTION

▶ Video Solution

$$\begin{aligned} \left| \cdot A^{-1} \cdot \operatorname{adj}(B^{-1}) \cdot \operatorname{adj}(2A^{-1}) \right| &= \frac{1}{|A|} \cdot \frac{1}{|B|^2} \cdot |2A^{-1}|^2 \\ &= \frac{1}{|A|} \cdot \frac{1}{|B|^2} \cdot \frac{64}{|A^2|^2} = \frac{64}{|A|^3 |B|^2} = \frac{64}{8 \times 9} = \frac{8}{9} \end{aligned}$$

Let $f(x) = (\sin(\tan^{-1}x) + \sin(\cot^{-1}x))^2 - 1$, $|x| < 1$ If

$\frac{dy}{dx} = \frac{1}{2} \frac{d}{dx} (\sin^{-1}(f(x)))$ and $y\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$, then $y\left(\frac{-1}{\sqrt{3}}\right)$ is equal to:

A

$$\frac{2\pi}{3}$$

3.73% marked this

B

$$-\frac{\pi}{6}$$

Correct
75.29% marked this

C

$$\frac{5\pi}{6}$$

17.75% marked this

D

$$\frac{\pi}{3}$$

3.24% marked this

TIME TAKEN

Others: 202 sec

TIME ALLOTTED

210 sec

DIFFICULTY

Hard

CHAPTER

Differential Equations

TOPIC

Variable Separable Form

DIDN'T ATTEMPT

66.23%

GOT IT RIGHT

75.29%

GOT IT WRONG

24.71%

SOLUTION

▶ Video Solution

Let $\tan^{-1}x = \theta$

$$\Rightarrow x = \tan\theta \Rightarrow \sin\theta = \frac{x}{\sqrt{1+x^2}}$$

$$\text{So, } y = \left(\frac{x}{\sqrt{1+x^2}} + \frac{1}{\sqrt{1+x^2}} \right)^2 - 1$$

$$\Rightarrow y = \frac{(x+1)^2}{1+x^2} - 1$$

$$\Rightarrow y = \frac{2x}{1+x^2} = f(x)$$

$$\frac{dy}{dx} = \frac{1}{2} \frac{d}{dx} (2\tan^{-1}x)$$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

$$\Rightarrow y = \tan^{-1}x + c$$

$$y\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6} \Rightarrow c = 0$$

$$\therefore y\left(\frac{-1}{\sqrt{3}}\right) = -\frac{\pi}{6}$$

If $f(\theta) = \frac{1 - \sin 2\theta + \cos 2\theta}{2 \cos 2\theta}$ then value of $f(11^\circ) \cdot f(34^\circ)$ equals

A $\frac{1}{2}$

Correct

73.12% marked this

B $\frac{3}{4}$

10.42% marked this

C $\frac{1}{4}$

7.95% marked this

D 1

8.52% marked this

EVALUATION

TIME TAKEN

Others: 162 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

Trigonometric Ratios & Identities

TOPIC

Sum and Difference & Multiple angle Formula

DIDN'T ATTEMPT

58.23%

GOT IT RIGHT

73.12%

GOT IT WRONG

26.88%

SOLUTION

▶ Video Solution

$$\begin{aligned} f(\theta) &= \frac{1 - \sin 2\theta + \cos 2\theta}{2 \cos 2\theta} = \frac{(\cos \theta - \sin \theta)^2 + (\cos^2 \theta - \sin^2 \theta)}{2(\cos \theta - \sin \theta)(\cos \theta + \sin \theta)} = \frac{(\cos \theta - \sin \theta) + (\cos \theta + \sin \theta)}{2(\cos \theta + \sin \theta)} \\ &= \frac{2 \cos \theta}{2(\cos \theta + \sin \theta)} = \frac{1}{1 + \tan \theta} \\ f(11^\circ) \cdot f(34^\circ) &= \frac{1}{(1 + \tan 11^\circ)} \cdot \frac{1}{(1 + \tan 34^\circ)} = \frac{1}{(1 + \tan 11^\circ)} \cdot \frac{1}{(1 + \tan(45^\circ - 11^\circ))} \\ &= \frac{1}{(1 + \tan 11^\circ)} \cdot \frac{1}{1 + \frac{1 - \tan 11^\circ}{1 + \tan 11^\circ}} = \frac{1}{(1 + \tan 11^\circ)} \cdot \frac{1 + \tan 11^\circ}{2} = \frac{1}{2} \end{aligned}$$

$A(27, -243, 81)$ is a point in space. B, C, D are images of A with respect to XY, YZ and ZX planes respectively. If the centroid of the triangle BCD is (α, β, γ) , then $\alpha + \beta + \gamma =$

A

-25

1.35% marked this

B

45

8.77% marked this

C

25

0.79% marked this

D

-45

Correct

89.09% marked this

EVALUATION

TIME TAKEN

TIME ALLOTTED

Others: 153 sec

150 sec

DIFFICULTY

Easy

CHAPTER

Three Dimensional Geometry

TOPIC

Basics of point in 3 dimension

DIDN'T ATTEMPT



44.08%

GOT IT RIGHT



89.09%

GOT IT WRONG



10.91%

SOLUTION

[▶ Video Solution](#)

Given point $A(27, -243, 81)$ in space, now images of point A in xyh plane, yzh plane and zx-planes are B $(27, -243, -81)$, $C(-27, -243, 81)$ and $D(27, 243, 81)$ respectively.

Hence centroid of the triangle BCD is $= (9, -81, 27) \equiv (\alpha, \beta, \gamma)$ hence,
 $\alpha + \beta + \gamma = -45$

If $f(x) = \int 2^{\frac{\sin x}{\log_e 2}} \left(\frac{5 \cos x + \cos^3 x}{(2 - \sin x) \sqrt{3 + \cos^2 x}} \right) dx$ and $f(0) = 1$ then value of $f(\pi/2)$ equals

A $\sqrt{3}e$
Correct
 54.85% marked this

B $\frac{\sqrt{3}e}{2}$
 31.07% marked this

C $\frac{3e}{2}$
 10.68% marked this

D $3e$
 3.40% marked this

DIFFICULTY
 Hard

DIDN'T ATTEMPT ⦿
 83.06%

TIME TAKEN
 Others: 174 sec

CHAPTER
 Indefinite Integration

GOT IT RIGHT ✔
 54.85%

TIME ALLOTTED
 240 sec

TOPIC
 Integration by Parts

GOT IT WRONG ✖
 45.15%

SOLUTION

[▶ Video Solution](#)

$$\int e^{\sin x} \left(\frac{5 + \cos^2 x}{(2 - \sin x) \sqrt{3 + \cos^2 x}} \right) \cos x dx$$

put $\sin x = t$

$$\int e^t \left(\frac{6 - t^2}{(2 - t) \sqrt{4 - t^2}} \right) dt$$

$$\int e^t \left(\frac{4 - t^2}{(2 - t) \sqrt{4 - t^2}} + \frac{2}{(2 - t) \sqrt{4 - t^2}} \right) dt$$

$$\int e^t \left(\sqrt{\frac{2+t}{2-t}} + \frac{2}{(2-t)^{3/2} (2+t)^{1/2}} \right) dt$$

$$\text{If } g(t) = \sqrt{\frac{2+t}{2-t}}, g'(t) = \frac{2}{(2-t)^{3/2} (2+t)^{1/2}}$$

$$e^t \sqrt{\frac{2+t}{2-t}} + c$$

$$f(x) = e^{\sin x} \sqrt{\frac{2 + \sin x}{2 - \sin x}} + c$$

$$f(0) = 1 \Rightarrow c = 0$$

$$f(x) = e^{\sin x} \sqrt{\frac{2 + \sin x}{2 - \sin x}}$$

$$f\left(\frac{\pi}{2}\right) = \sqrt{3}e$$

Let g_n be the n^{th} term of the geometric progression of positive numbers. If $\sum_{n=1}^{100} g_{2n} = \frac{10}{3}$ and $\sum_{n=1}^{100} g_{2n-1} = \frac{5}{9}$ then the common ratio of geometric progression, is

- A** 2
3.07% marked this
- B** 4
3.38% marked this
- C** 6
Correct
92.97% marked this
- D** 8
0.58% marked this

TIME TAKEN

Others: 184 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Easy

CHAPTER

Sequences and Series

TOPIC

Geometric Progression

DIDN'T ATTEMPT

40.90%

GOT IT RIGHT

92.97%

GOT IT WRONG

7.03%

SOLUTION

[▶ Video Solution](#)

Let first term and common ratio be a and r respectively.

$$\text{Now, } \sum_{n=1}^{100} g_{2n} = ar + ar^3 + ar^5 + \dots + ar^{199} = \frac{10}{3} \dots(1)$$

$$\text{Also, } \sum_{n=1}^{100} g_{2n-1} = a + ar^2 + ar^4 + \dots + a^{198} = \frac{5}{9} \dots(2)$$

$$\text{Now, } \frac{\text{Equation (1)}}{\text{Equation (2)}} \Rightarrow \frac{r(a + ar^2 + \dots + ar^{198})}{(a + ar^2 + \dots + ar^{198})} = \frac{10/3}{5/9} \Rightarrow r = 6$$

$$10 \int_0^\pi x \left(\sin^2(\sin x) + \cos^2(\cos x) \right) dx =$$

Round off ans to nearest integer (given $\pi^2 = 9.8$)

Your Answer

Correct Answer

49

TIME TAKEN

Others: 104 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Medium

CHAPTER

Definite Integration

TOPIC

Properties of Definite Integration

DIDN'T ATTEMPT 

63.03%

GOT IT RIGHT 

54.80%

GOT IT WRONG 

45.20%

SOLUTION

 Video Solution

$$10 \int_0^\pi x \left(\sin^2(\sin x) + \cos^2(\cos x) \right) dx =$$

$$I = \int_0^\pi x \left(\sin^2(\sin x) + \cos^2(\cos x) \right) dx$$

$$\Rightarrow I = \int_0^\pi (\pi - x) \left(\sin^2(\sin x) + \cos^2(\cos x) \right) dx$$

$$\text{Adding } 2I = \pi \int_0^\pi \left(\sin^2(\sin x) + \cos^2(\cos x) \right) dx$$

$$2I = 2\pi \int_0^{\pi/2} \left(\sin^2(\sin x) + \cos^2(\cos x) \right) dx$$

$$I = \pi \int_0^{\pi/2} \left(\sin^2(\sin x) + \cos^2(\cos x) \right) dx$$

$$\text{Also } I = \pi \int_0^{\pi/2} \left[\sin^2(\cos x) + \cos^2(\sin x) \right] dx$$

$$\text{Adding } 2I = \pi \int_0^{\pi/2} 2 dx = 2\pi \cdot \frac{\pi}{2} = \pi^2$$

$$\Rightarrow I = \frac{\pi^2}{2}$$

$$10 I = 5\pi^2 = 49$$

$OABC$ is a tetrahedron in with O as the origin and position vectors of points A, B, C as $\hat{i} + 2\hat{j} + 3\hat{k}$, $2\hat{i} + \alpha\hat{j} + \hat{k}$ and $\hat{i} + 3\hat{j} + 2\hat{k}$ respectively, then the integral value of α to have shortest distance between $\overrightarrow{OA}$ & $\overrightarrow{BC}$ as $\sqrt{\frac{3}{2}}$, is

Your Answer

Correct Answer

3

TIME TAKEN

Others: 219 sec

TIME ALLOTTED

210 sec

DIFFICULTY

Hard

CHAPTER

Three Dimensional Geometry

TOPIC

Line in Space

DIDN'T ATTEMPT 

62.60%

GOT IT RIGHT 

87.47%

GOT IT WRONG 

12.53%

SOLUTION

[▶ Video Solution](#)

Unit vector ($\hat{n}$) perpendicular to $\overrightarrow{OA}$ and $\overrightarrow{CB}$

$$\hat{n} = \frac{\overrightarrow{CB} \times \overrightarrow{OA}}{|\overrightarrow{CB} \times \overrightarrow{OA}|} = \frac{(3\alpha-7)\hat{i} - 4\hat{j} + (5-\alpha)\hat{k}}{\sqrt{(3\alpha-7)^2 + 16 + (5-\alpha)^2}}$$

$$\text{As } \overrightarrow{CB} \times \overrightarrow{OA} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \alpha-3 & -1 \\ 1 & 2 & 3 \end{vmatrix} = |\overrightarrow{BA} \cdot \hat{n}| = \left| \frac{(7-3\alpha) - 4(2-\alpha) + 2(5-\alpha)}{\sqrt{(3\alpha-7)^2 + 16 + (5-\alpha)^2}} \right|$$

$$\text{Shortest distance} = |\overrightarrow{BA} \cdot \hat{n}|$$

In a triangle ABC, $2\hat{i} + \hat{j}$, $5\hat{i} + \hat{j}$ & $2\hat{i} + 7\hat{j}$ be the position vectors of points A, B & C respectively. Let $\overrightarrow{AD}$, $\overrightarrow{AB}$ & $\overrightarrow{AC}$ are non-coplanar vectors. If 'M' is minimum integral value of $(\overrightarrow{AD})^2 + (\overrightarrow{BD})^2 + (\overrightarrow{CD})^2$ then sum of digits of M is

Your Answer

Correct Answer

4

TIME TAKEN

Others: 78 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Medium

CHAPTER

Vector Algebra

TOPIC

Product of 2 vectors

DIDN'T ATTEMPT 

69.45%

GOT IT RIGHT 

56.63%

GOT IT WRONG 

43.37%

SOLUTION

[▶ Video Solution](#)

Let $D = (x, y, z)$

$$\begin{aligned}
 \therefore (AD)^2 + (BD)^2 + (CD)^2 &= (x-1)^2 + (y-1)^2 + z^2 + (x-5)^2 \\
 &+ (y-1)^2 + z^2 + (x-2)^2 + (y-7)^2 + z^2 \\
 &= 3x^2 - 18x + 3y^2 - 18y + 84 + 3z^2 \\
 &= 3(x-3)^2 + 3(y-3)^2 + 30 + 3z^2 \\
 \therefore M &= 31
 \end{aligned}$$

The constant term in the expansion of $\left(1 + x + \frac{2}{x}\right)^6$ is

Your Answer

Correct Answer

581

EVALUATION

TIME TAKEN

Others: 110 sec

TIME ALLOTTED

210 sec

DIFFICULTY

Hard

CHAPTER

Binomial Theorem

TOPIC

Multinomial Theorem

DIDN'T ATTEMPT 

40.30%

GOT IT RIGHT 

62.09%

GOT IT WRONG 

37.91%

SOLUTION

[▶ Video Solution](#)

$$\left(1 + x + \frac{2}{x}\right)^6 = 1 + \binom{6}{1}\left(x + \frac{2}{x}\right) + \binom{6}{2}\left(x + \frac{2}{x}\right)^2 + \dots + \binom{6}{6}\left(x + \frac{2}{x}\right)^6$$

$$\text{The constant term is } 1 + \binom{6}{2}\binom{2}{1}2^1 + \binom{6}{4}\binom{4}{2}2^2 + \binom{6}{6}\binom{6}{3}2^3.$$

$$= 1 + 60 + 360 + 160 = 581.$$

If $y - \cos x \frac{dy}{dx} = y^2(1 - \sin x) \cos x$, $y(0) = 1$, then $y\left(\frac{\pi}{3}\right) =$

Your Answer

Correct Answer

2

TIME TAKEN

Others: 420 sec

TIME ALLOTTED

210 sec

DIFFICULTY

Medium

CHAPTER

Differential Equations

TOPIC

Linear Differential Equation

DIDN'T ATTEMPT 

66.96%

GOT IT RIGHT 

89.76%

GOT IT WRONG 

10.24%

SOLUTION

[▶ Video Solution](#)

Dividing the given d.e by $y^2 \cos x$, $-\frac{1}{y^2} \frac{dy}{dx} + \sec x \cdot \frac{1}{y} = 1 - \sin x$

$$\frac{d}{dx} \left(\frac{1}{y} \right) + \sec x \cdot \left(\frac{1}{y} \right) = 1 - \sin x$$

Linear d.e with I.F. = $\exp \int \sec x dx = \sec x + \tan x$

$$\text{Solution, } \frac{\sec x + \tan x}{y} = \int \frac{(1 - \sin x)(1 + \sin x) dx}{\cos x} = \int \cos x dx = \sin x + c$$

$$x = 0, y = 1 \Rightarrow c = 1$$

$$\therefore y = \frac{\sec x + \tan x}{\sin x + 1} = \sec x, y\left(\frac{\pi}{3}\right) = 2.$$